SUPPLEMENT TO THE q-FOULKES PROJECT: ADDITIONAL RESULTS AND CERTIFICATES

SUVRIT SRA*

ABSTRACT. This supplement collects further results on Bergeron's q-analog of Foulkes' conjecture that support the three papers of the series. In the Gaussian Newton expansions of Paper III, the two highest coefficients are Schur positive in large depths. For hook shapes, an explicit positive model of the coefficients gives a shifted degree comparison and rules out the simplest couplings for an unshifted one. A classification of the kernels of exchange operators gives their graded characters and ranks. Finally, capacity, determinant, and quantum-permanent criteria verify the expansion hypothesis of Paper III on individual isotypic blocks; the hypothesis remains open in general.

CONTENTS

Introduction 2

Part 1. Gaussian Newton expansions 2
S1. Schur functors and positive Gaussian expansions 2
S2. The highest Newton coefficient in every depth 4
S3. The second-highest Newton coefficient 7

Part 2. Hook coefficients and couplings 11
S4. Hook coefficients and matrix couplings 11

Part 3. Exchange operators and graded kernels 18
S5. The exchange gain and degree cutoffs 18
S6. Graded kernels, incremental loss, and transfers 20
S7. Positive averages over final cutoffs 34

Part 4. Polarization and the expansion hypothesis 36
S8. Quantitative power polarization 36
S9. The current space and the expansion hypothesis 36
S10. Certificates for the expansion hypothesis 37
S11. Exact block calculations 40
S12. Exact checks 47
References 47

2020 *Mathematics Subject Classification.* Primary 05E05, 20C30; Secondary 14C05, 15A15.

Key words and phrases. q-Foulkes conjecture, Gaussian Newton expansion, hook coefficients, exchange operators, current-space certificates.

* TU Munich, Dept. of Mathematics, School of CIT; Garching, Germany. Email: s.sra@tum.de.

Introduction

The three papers of the series study Bergeron's q-analog of Foulkes' conjecture through a fusion inequality for Macdonald plethysms (Sra, 2026a), Schur positivity with high probability (Sra, 2026b), and canonical maps with a graded inner factor (Sra, 2026c). The results collected here extend their methods but are not needed for their main theorems. The Gaussian Newton expansions and the current space come from Paper III, the hook models refine the shifted comparison of Paper I, and the degree comparisons use the filtered formulation of Paper II. The expansion hypothesis of Paper III remains open. References with an S prefix belong to this supplement; references to the three papers carry their paper numbers.

We use the conventions of Paper III, Section 1.1. In particular, plethysm acts on every grading variable through the Adams operations, tensor permutations carry no Koszul signs, and

$$H_n(X;q) = (q;q)_n h_n[X/(1-q)], \qquad P_{a,b} = H_b[H_a], \quad Q_{a,b} = H_a[H_b], \quad F_{a,b} = \frac{P_{a,b} - Q_{a,b}}{1-q}.$$

Unless stated otherwise, $a, b \geq 2$ are integers and $N = ab$. Write $c_n = \binom{n}{2}$, $\varepsilon_n = n \bmod 2$, and $n(\lambda) = \sum_i (i-1)\lambda_i$. The symbols $[n]_q$ and $\begin{bmatrix} n \\ k \end{bmatrix}_q$ denote the q-integer and Gaussian binomial coefficient. For $\lambda \vdash N$, write S^λ for the Specht module, χ^λ for its character, and $d_\lambda = \dim S^\lambda$. Its height and depth are $\ell(\lambda)$ and $N - \lambda_1$, and the involution ω sends s_λ to $s_{\lambda'}$. We write $K\{s_\lambda\}$ for the symmetric functions with Schur coefficients in a semiring K, and $\succeq_s 0$ for coefficientwise Schur positivity. As in Paper II, $P_{a,b}$ is the target character and $Q_{a,b}$ the source character in the filtered comparison. For maps running from $\mathcal{P}$ to $\mathcal{Q}$, we use domain and codomain.

Part 1. Gaussian Newton expansions

Paper III proves positivity at bounded depth by computing every coefficient $A^{(B)}_{a,r,j}$ of the expansion

$$\sum\nolimits_{\nu \vdash r} [s_{(ab-r,\nu)}] F_{a,b}\, s_\nu(Y) = \sum\nolimits_{j=0}^{r} A^{(B)}_{a,r,j}(Y;q) \begin{bmatrix} b-B \\ j \end{bmatrix}_q, \qquad b \geq B \geq a, \quad aB \geq 2r,$$

where the restrictions $\mathcal{R}^{(r)}_{a,b} = h^{\perp}_{ab-r} P_{a,b}$ and their triangular recovery are as in Paper III, Section 3. Computation cannot reach every depth, so this part proves positivity structurally where a uniform description exists. Each orientation alone has a positive expansion (Section S1). The difficulty lies in the difference of the two orientations, which we control at the two highest indices $j = r$ and $j = r-1$.

S1. Schur functors and positive Gaussian expansions

A Schur functor applied to a family with a positive Gaussian expansion again has one. This closure gives the source restriction positive Gaussian coefficients, and a direct expansion gives them for the target, so each orientation is positive before the two are compared. The proof of closure lifts Gaussian polynomials to exterior powers and then uses principal specialization of Schur functions.

After the lift, the Schur functor of the family is a sum of principal specializations $s_\lambda[X_n]$, where $X_n = 1 + q + \cdots + q^{n-1}$, multiplied by Schur functions of Y and by powers of q whose exponents may be negative but are at least $-n(\lambda)$. The next lemma expands $s_\lambda[X_n]$ in Gaussian binomials with coefficients divisible by $q^{n(\lambda)}$, and this divisibility absorbs these powers. For a standard tableau T of size d, let $\mathrm{Des}(T)$ be the set of i such that $i+1$ lies in a lower row of T than i, let $d(T) = |\mathrm{Des}(T)|$, and put $c(T) = \sum_{i \in \mathrm{Des}(T)} (d-i)$.

Lemma S1.1 (A positive principal specialization)**.** *For $\lambda \vdash d > 0$,*

$$s_\lambda[X_n] = \sum_{j=0}^{d} B_{\lambda,j}(q) \begin{bmatrix} n \\ j \end{bmatrix}_q, \qquad B_{\lambda,j} = \sum_{T \in \mathrm{SYT}(\lambda)} q^{c(T)+j(j-1-d(T))} \begin{bmatrix} d-1-d(T) \\ d-j \end{bmatrix}_q. \tag{S1.1}$$

Terms with vanishing Gaussian binomial are omitted. Every coefficient $B_{\lambda,j}$ belongs to $q^{n(\lambda)}\mathbb{N}[q]$.

Proof. Standardization expresses a semistandard tableau as a standard tableau T together with

$$0 \le x_1 \le \cdots \le x_d < n, \qquad x_i < x_{i+1} \quad (i \in \mathrm{Des}(T)).$$

Subtracting the number of earlier descents from each x_i gives

$$s_\lambda[X_n] = \sum_T q^{c(T)} \begin{bmatrix} n+d-1-d(T) \\ d \end{bmatrix}_q.$$

The identity $e_j[X_n] = q^{\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_q$ and the alphabet splitting $X_{n+c} = X_c + q^c X_n$ give

$$\begin{bmatrix} n+c \\ d \end{bmatrix}_q = \sum_j q^{j(c-d+j)} \begin{bmatrix} c \\ d-j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q.$$

Taking $c = d-1-d(T)$ proves (S1.1). A nonzero summand requires $j \ge d(T)+1$, so its extra exponent is nonnegative. The minimal admissible sequence for T is itself semistandard; an entry in row i is at least $i-1$. Thus $c(T) \ge n(\lambda)$. □

Theorem S1.2 (Closure under Schur functors)**.** *Suppose*

$$E_n(Y;q) = \sum_{j=0}^{J} \begin{bmatrix} n \\ j \end{bmatrix}_q \Phi_j(Y;q), \qquad \Phi_j \in \mathbb{N}[q]\{s_\lambda(Y)\}.$$

For every partition μ, the family $s_\mu[E_n]$ has a finite Gaussian Newton expansion with Schur-positive coefficients over $\mathbb{N}[q]$. The same conclusion holds for every nonnegative q-polynomial combination of Schur functors, including H_m.

Proof. Realize Φ_j as a nonnegatively graded polynomial representation $\mathcal{E}_j$. On an auxiliary vector space U, form

$$\mathcal{W}(U,Y) = \bigoplus_j q^{-\binom{j}{2}} \Lambda^j U \otimes \mathcal{E}_j(Y).$$

Although this representation has a Laurent grading, its specialization $U = X_n$ is nonnegatively graded and has character E_n. Semisimplicity in characteristic zero gives a finite decomposition

$$S_\mu \mathcal{W} = \bigoplus_{\lambda,\nu,d} m_{\lambda,\nu,d} q^d S_\lambda(U) \otimes S_\nu(Y), \qquad m_{\lambda,\nu,d} \in \mathbb{N}.$$

Every occurring summand satisfies $d + n(\lambda) \ge 0$. Indeed, specializing $U = X_n$ for sufficiently large n gives a Schur functor of a nonnegatively graded space, whereas the summand has lowest degree $d+n(\lambda)$; a negative degree cannot cancel among positive characters. Lemma S1.1 therefore expands every $q^d s_\lambda[X_n]$ with coefficients in $\mathbb{N}[q]$. The constant Schur summand contributes only at Gaussian index zero. Substitution in the decomposition proves the assertion. □

Let $aB \ge r$, so that the restrictions $\mathcal{R}^{(r)}_{a,b} = h^{\perp}_{ab-r} H_b[H_a]$ are defined for all $b \ge B$. Then $\mathcal{R}^{(r)}_{a,B+n}$ and $\mathcal{R}^{(r)}_{B+n,a}$ have positive Gaussian expansions in n with indices at most r. For the source we use

$$H_{B+n}[1+Y] = \sum_{t \ge 0} \begin{bmatrix} B+n \\ t \end{bmatrix}_q H_t[Y], \qquad \begin{bmatrix} B+n \\ t \end{bmatrix}_q = \sum_j q^{j(B-t+j)} \begin{bmatrix} B \\ t-j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q, \tag{S1.2}$$

truncated at Y-degree r, together with Theorem S1.2 applied to H_a. Every surviving term has $t-j \le B$, so its shift is nonnegative. The auxiliary degree in the proof of closure is at most the Y-degree; Lemma S1.1 then bounds the Newton index by r.

For the target, set $\mathcal{Z}_a = H_a[1+Y]-1$. Its minimum degree is one, and

$$\mathcal{R}^{(r)}_{a,B+n} = \sum\nolimits_{t=0}^{r} \begin{bmatrix} B+n \\ t \end{bmatrix}_q [\deg_Y = r] H_t[\mathcal{Z}_a].$$

The bracketed characters are positive, so (S1.2) applies termwise. At the highest index the target coefficient is $q^{Br} H_r[[a]_q Y]$.

Closure concerns the two restrictions separately; their difference and the signed triangular recovery need other arguments. Even an Adams operation can destroy a positive Gaussian expansion, since $[n]_{q^2} = [n]_q + (q^2-q)\begin{bmatrix} n \\ 2 \end{bmatrix}_q$.

S2. The highest Newton coefficient in every depth

The highest coefficient $A^{(B)}_{a,r,r}$ is Schur-positive at every depth $r \ge 2a-1$, and Section S3 uses this positivity to reach a finite baseline at the next index. At the highest index, triangular recovery contributes no lower-depth terms. The remaining comparison admits matching occupancy expansions, in which the target and source have the same ungraded character and their graded supports separate once the depth is large enough.

Theorem S2.1 compares a target character $C_{a,r}$ with a source character $D_{a,r}$, which sums over the ways r selected labels occupy the a blocks of the source. Let $*$ denote the internal Kronecker product. For $\lambda = (1^{m_1}2^{m_2}\cdots) \vdash r$ with $\ell(\lambda) \le a$, put

$$m_0 = a - \ell(\lambda), \qquad e(\lambda) = c_r - \sum\nolimits_{j\ge 1} m_j c_j = \sum\nolimits_{i<j} \lambda_i \lambda_j.$$

With the Gaussian multinomials of Paper III, Section 3.1, define

$$U_\lambda = \prod\nolimits_{j \text{ even}} H_{m_j}[h_j] \prod\nolimits_{j \text{ odd}} (\omega H_{m_j})[h_j],$$

$$C_{a,r} = h_r[[a]_q Y], \qquad D_{a,r} = \sum_{\substack{\lambda \vdash r \\ \ell(\lambda) \le a}} q^{e(\lambda)} \begin{bmatrix} a \\ m_0, m_1, \ldots \end{bmatrix}_q U_\lambda. \tag{S2.1}$$

The involution acts on the outer H_{m_j} before plethysm. Both $C_{a,r}$ and $D_{a,r}$ are positive characters; set $C_{a,0} = D_{a,0} = 1$.

Theorem S2.1 (Highest Gaussian index)**.** *Let $B \ge a$ and $aB \ge 2r$. In the expansion of Paper III, Equation (3.4), the coefficient $A^{(B)}_{a,r,r}$ is Schur-positive over $\mathbb{N}[q]$ whenever $r \ge 2a-1$. The same conclusion holds at every depth $r \ge 0$ for $1 \le a \le 7$. More precisely,*

$$A^{(B)}_{a,r,r} = q^{rB} H_r * \frac{C_{a,r} - D_{a,r}}{1-q}, \tag{S2.2}$$

and the quotient before the internal product is itself Schur-positive in the stated ranges.

Both orientations in (S2.2) share the internal product with H_r, the graded Frobenius character of the coinvariant algebra, so the comparison reduces to the supports of the occupancy terms. For $a \le 7$, the finitely many depths $r < 2a-1$ have exact certificates in Section S2.2.

S2.1. **The highest Gaussian index.** Only the leading coefficients in $z = q^b$ reach the top Gaussian index. In both orientations they are sums over occupancies $\lambda \vdash r$ with the same Gaussian multinomial weights and the same ungraded character h_λ, so Theorem S2.1 reduces to comparing graded supports occupancy by occupancy.

S2.1.1. *The common coinvariant factor.* To identify the contribution of the source to (S2.2), put $z = q^b$. The source restriction is $\mathcal{R}^{(r)}_{b,a} = [\deg_Y = r]\, H_a[\sum_{j\ge 0} \begin{bmatrix} b \\ j \end{bmatrix}_q H_j]$, because $h^{\perp}_{N-r} f$ is the part of $f[1+Y]$ of degree r in Y and $H_b[1+Y] = \sum_j \begin{bmatrix} b \\ j \end{bmatrix}_q H_j[Y]$. The leading coefficient of $\begin{bmatrix} b \\ j \end{bmatrix}_q$ is $(-1)^j q^{-c_j}/(q;q)_j$, so at leading order the alphabet $\begin{bmatrix} b \\ j \end{bmatrix}_q H_j$ is replaced by

$$K_j = (-1)^j q^{-c_j} h_j[Y/(1-q)], \qquad K_0 = 1.$$

The leading coefficient of $\begin{bmatrix} b-B \\ r \end{bmatrix}_q$ is $(-1)^r q^{-rB-c_r}/(q;q)_r$, so the source coefficient over q^{rB} is

$$(-1)^r q^{c_r} (q;q)_r [\deg_Y = r] H_a\left[\sum\nolimits_{j\ge 0} K_j\right]. \tag{S2.3}$$

Adams operations fix the integer sign $(-1)^j$ and act on the grading by $q \mapsto q^t$.

In (S2.3) every factor comes from the alphabet $Y/(1-q)$, so a power sum $p_\rho(Y)$ carries the denominator $\prod_{t\in\rho}(1-q^t)$. Multiplication by $(q;q)_r$ leaves the graded trace $\kappa_r(\rho;q)$ of the coinvariant algebra in Paper III, Equation (3.2). Thus this multiplication is the internal product with H_r. We expand the other factor by occupancies through $H_a[X+X'] = \sum_k \begin{bmatrix} a \\ k \end{bmatrix}_q H_k[X] H_{a-k}[X']$ and obtain

$$(-1)^r \sum\nolimits_\lambda q^{e(\lambda)} \begin{bmatrix} a \\ m_0, m_1, \dots \end{bmatrix}_q \prod\nolimits_j H_{m_j}[(-1)^j h_j].$$

For odd j, use $H_m[-X] = (-1)^m (\omega H_m)[X]$. Since $\sum_{j \text{ odd}} m_j \equiv r \pmod 2$, the signs cancel and leave $D_{a,r}$. In the target only the occupancy 1^r has z-degree r; its r occupied blocks hold one selected label each, and it gives $H_r[[a]_q Y] = H_r * C_{a,r}$. Lower restrictions have degree below r in z, so they do not affect this coefficient. Subtraction and division by $1-q$ prove (S2.2).

S2.1.2. *Hall–Littlewood expansion and graded supports.* To compare $C_{a,r}$ with $D_{a,r}$ occupancy by occupancy, we expand $C_{a,r}$ in transformed Hall–Littlewood functions, in the convention

$$Q'_\lambda(Y;q) = \sum\nolimits_\nu K_{\nu\lambda}(q) s_\nu(Y), \qquad Q'_{11} = s_{11} + q s_2.$$

Their coefficients are nonnegative, their degrees are at most $n(\lambda)$, and $Q'_\lambda(Y;1) = h_\lambda(Y)$ (Macdonald, 1995, Chapter III). The Hall–Littlewood Cauchy identity reads $h_r[XY] = \sum_{\lambda\vdash r} P_\lambda(X;q) Q'_\lambda(Y;q)$. At the alphabet $X = 1 + q + \cdots + q^{a-1}$ it gives

$$C_{a,r} = \sum_{\substack{\lambda\vdash r \\ \ell(\lambda)\le a}} q^{n(\lambda)} \begin{bmatrix} a \\ m_0, m_1, \dots \end{bmatrix}_q Q'_\lambda(Y;q). \tag{S2.4}$$

To evaluate $P_\lambda(1, q, \dots, q^{a-1}; q)$, pad λ to a parts and use

$$P_\lambda(x_1, \dots, x_a; q) = \frac{1}{\prod_{j\ge 0}[m_j]_q!} \sum\nolimits_{\sigma\in S_a} \sigma\left(x_1^{\lambda_1} \cdots x_a^{\lambda_a} \prod\nolimits_{i<j} \frac{x_i - q x_j}{x_i - x_j} \right).$$

At $(x_1, \dots, x_a) = (1, q, \dots, q^{a-1})$, an ordering placing q^{t+1} before q^t has a zero numerator factor. Only the increasing ordering survives. Its monomial is $q^{n(\lambda)}$ and its Vandermonde ratio is $[a]_q!$. The denominator includes repeated zero parts, yielding the multinomial in (S2.4).

Each occupancy carries the same Gaussian multinomial in (S2.4) and (S2.1), so it suffices to compare $q^{n(\lambda)} Q'_\lambda$ with $q^{e(\lambda)} U_\lambda$. Both specialize to h_λ at $q = 1$. Every target degree is at most $2n(\lambda)$, and every source degree is at least $e(\lambda)$. If $e(\lambda) \ge 2n(\lambda)$, pair the copies of each irreducible in these

two ungraded characters. A pair with degrees $u \le v$ contributes $(q^u - q^v)/(1-q) = q^u[v-u]_q$, so every occupancy λ satisfies

$$e(\lambda) \ge 2n(\lambda) \quad \Longrightarrow \quad \frac{q^{n(\lambda)}Q'_\lambda - q^{e(\lambda)}U_\lambda}{1-q} \in \mathbb{N}[q]\{s_\nu\}. \tag{S2.5}$$

It remains to decide which occupancies satisfy $e(\lambda) \ge 2n(\lambda)$.

Lemma S2.2 (Support threshold)**.** *If $\lambda \vdash r$ has length m and $r \ge 2m-1$, then $e(\lambda) \ge 2n(\lambda)$. For $m \ge 2$, the threshold is sharp: some $\lambda \vdash 2m-2$ of length m has $e(\lambda) < 2n(\lambda)$.*

Proof. The difference is

$$e(\lambda) - 2n(\lambda) = \sum_{j=2}^{m} \lambda_j \sum_{i<j} (\lambda_i - 2). \tag{S2.6}$$

Let k be the number of trailing ones and write the other parts as $2+\delta_i$, with $\delta_i \ge 0$ and $\Delta = \sum_i \delta_i = r - 2m + k$. The contributions from parts at least two are nonnegative. The trailing ones contribute $k\Delta - \binom{k}{2}$. If $k > 0$, the hypothesis gives $\Delta \ge k-1$, so this contribution is nonnegative too. For $k = 0$, all terms are already nonnegative. For $m \ge 2$ at size $2m-2$, the partition $(2^{m-2}, 1, 1)$ has $e(\lambda) - 2n(\lambda) = -1$. □

Proof of Theorem S2.1. If $r \ge 2a-1$, Lemma S2.2 applies to every occupancy in (S2.4). Summing (S2.5) and using (S2.2) proves positivity. For $a \le 7$, the condition $r \ge 2a-1$ leaves only $r \le 12$. The exact certificates of Section S2.2 prove the remaining range. At depth zero the coefficient vanishes, since $C_{a,0} = D_{a,0} = 1$. □

S2.2. **Exact certificates for the highest index.** For $a \le 7$ the proof of Theorem S2.1 leaves only $r \le 12$. The inner size a factors out once the characters in (S2.1) and (S2.4) are grouped by the number m of occupied blocks,

$$C_{a,r} = \sum_m \begin{bmatrix} a \\ m \end{bmatrix}_q \Psi_{m,r}, \qquad D_{a,r} = \sum_m \begin{bmatrix} a \\ m \end{bmatrix}_q \Phi_{m,r},$$

where

$$\Psi_{m,r} = \sum_{\substack{\lambda \vdash r \\ \ell(\lambda) = m}} \begin{bmatrix} m \\ m_1, m_2, \dots \end{bmatrix}_q q^{n(\lambda)} Q'_\lambda,$$

$$\Phi_{m,r} = \sum_{\substack{\lambda \vdash r \\ \ell(\lambda) = m}} \begin{bmatrix} m \\ m_1, m_2, \dots \end{bmatrix}_q q^{e(\lambda)} U_\lambda.$$

For $1 \le r \le 12$ and $1 \le m \le \min(7, r)$, every Schur coefficient of $(\Psi_{m,r} - \Phi_{m,r})/(1-q)$ is a polynomial with nonnegative integer coefficients; the 63 pairs (m, r) give 1,829 such polynomials with 44,382 coefficients.

These polynomials need no Hall–Littlewood computation. We compute $C_{a,r} = h_r[[a]_q Y]$ for $a = 0, \dots, m$ from power sums, recover $\Psi_{m,r} = C_{m,r} - \sum_{j<m} \begin{bmatrix} m \\ j \end{bmatrix}_q \Psi_{j,r}$, compute $\Phi_{m,r}$ from the occupancy formula, and divide each Schur coefficient of the difference by $1-q$, checking for zero remainder. The same computation gives all 271 Schur coefficients of $C_{7,r} - D_{7,r}$ for $1 \le r \le 12$.

A second computation proves the same range occupancy by occupancy. For $\lambda = 1^m$ the two terms of (S2.5) coincide, because $Q'_{1^m} = \omega H_m$ and $e(1^m) = n(1^m)$. The other occupancies of length at most seven with $e(\lambda) < 2n(\lambda)$ are

$$(2^t, 1^s) \quad (1 \le t \le 5,\ 2 \le s \le 7-t),$$
$$(3,1^4),\ (3,1^5),\ (3,1^6),\ (4,1^6),\ (3,2,1^5).$$

Lemma S2.2 restricts their sizes to at most twelve, and enumerating the partitions of those sizes gives exactly these twenty. For them, Q'_λ comes from the raising-operator formula

$$Q'_\lambda = \prod\nolimits_{i<j} \frac{1-R_{ij}}{1-qR_{ij}} h_\lambda, \qquad R_{ij}(\alpha) = \alpha + \mathbf{e}_i - \mathbf{e}_j,$$

with $h_\alpha = \prod_i h_{\alpha_i}$ and $h_j = 0$ for $j < 0$. Here $(1-R_{ij})/(1-qR_{ij}) = 1 + \sum_{t\geq 1}(q^t - q^{t-1})R_{ij}^t$, and R_{ij} moves a box from part j to part $i < j$. If the factors are expanded in decreasing order of j, part j never grows once its own factors are expanded, so terms in which it is negative can be discarded at once. All twenty quotients $(q^{n(\lambda)}Q'_\lambda - q^{e(\lambda)}U_\lambda)/(1-q)$ are Schur positive over $\mathbb{N}[q]$; together they have 558 Schur coefficients with 20,692 integer coefficients.

S2.3. **The highest coefficient at depth four.** At depth four the highest coefficient has an explicit factorization that exhibits its dependence on the inner size a. For $a \geq 2$,

$$A^{(a)}_{a,4,\nu,4} = q^{4a+e_\nu}[a-1]_q[a]_q S_\nu(q)\big(P_\nu(q) + q^2 C_\nu(q)[a-2]_q\big), \tag{S2.7}$$

where $A^{(a)}_{a,4,\nu,4} = [s_\nu]A^{(a)}_{a,4,4}$ whenever $a^2 \geq 8$; for smaller a, the formula refers to the polynomial continuation of the fixed-depth expression. In the table below, the last two columns list coefficients in ascending powers of q.

ν	e_ν	S_ν	P_ν	C_ν
(4)	2	1	$2,2,4,2,2$	$0,1,3,2,2$
$(3,1)$	1	$[3]_q$	$1,1,4,2,3,1$	$0,2,2,1,3$
$(2,2)$	2	$1+q^2$	$2,2,4,2,2$	$2,2,2,1,1$
$(2,1,1)$	1	$[3]_q$	$0,1,3,2,4,1,1$	$1,0,3,2,1,1$
(1^4)	2	1	$0,0,2,2,4,2,2$	$1,1,2,1,2,1$

By (S2.2), $A^{(a)}_{a,4,4} = q^{4a}H_4 * (C_{a,4} - D_{a,4})/(1-q)$, and its five occupancy terms make each Schur coefficient a polynomial in q^a. After removing $q^{4a+e_\nu}[a-1]_q[a]_q S_\nu$, the result is $(D_\nu - q^a C_\nu)/(1-q)$ with $D_\nu = (1-q)P_\nu + q^2 C_\nu$. This quotient equals $P_\nu + q^2 C_\nu[a-2]_q$, which proves (S2.7). These five identities are checked formally in $\mathbb{Q}(q)[q^a]$.

S3. The second-highest Newton coefficient

The next index includes one lower-depth term from triangular recovery, namely h_1 times the highest coefficient at depth $r-1$. In the limit of large baseline, this term enters a correction that is positive by a tableau injection, and the occupancy comparison of Section S2 holds above a stronger support threshold. A degree estimate then gives a finite baseline.

Theorem S3.1 (Second-highest Gaussian index)**.** *For $a \in \{6,7\}$, every $r \geq 2$, and every integer $B \geq r(r-1) + \binom{a}{2} - 1$, the coefficient $A^{(B)}_{a,r,r-1}$ is Schur-positive over $\mathbb{N}[q]$. More generally, the conclusion holds for*

$$a \geq 5, \qquad r \geq 2a+3, \qquad B \geq r(r-1) + \binom{a}{2} - 1.$$

Positivity propagates to larger baselines because

$$A^{(B+1)}_{a,r,r-1} = q^{r-1}A^{(B)}_{a,r,r-1} + A^{(B)}_{a,r,r}.$$

S3.1. **The second-highest Gaussian index.** The next coefficient depends on the baseline beyond a monomial factor, because the source contributes a subleading term in $z = q^b$ and triangular recovery contributes the top coefficient at depth $r-1$. Its positivity is therefore proved first at $w = 0$, the formal limit of $w = q^B$ as $B \to \infty$, where it is again an occupancy comparison, now with one occupied block marked.

S3.1.1. *The exact coefficient including recovery.* To pass to the limit, we need an exact formula in which the baseline enters only through $w = q^B$ and a monomial factor. With the occupancy multiplicities of (S2.1), put

$$\Theta_m^{(j)} = \begin{cases} H_m, & j \text{ even}, \\ \omega H_m, & j \text{ odd}. \end{cases}$$

The character obtained by marking an occupied block is

$$V_\lambda = \sum_{j:m_j>0} [j]_q [m_j]_q \, h_j \Theta_{m_j-1}^{(j)}[h_j] \prod_{t\neq j} \Theta_{m_t}^{(t)}[h_t],$$

$$W_{a,r} = \sum_{\substack{\lambda\vdash r \\ \ell(\lambda)\le a}} q^{e(\lambda)} \begin{bmatrix} a \\ m_0, m_1, \ldots \end{bmatrix}_q V_\lambda. \tag{S3.1}$$

At $q = 1$, the marked character is $V_\lambda = r h_\lambda$. Define

$$\begin{aligned} N_{a,r}(w) = {} & [r]_q C_{a,r} - W_{a,r} - w[r]_q (C_{a,r} - D_{a,r}) \\ & + q^{r-1}(1-q^2) \begin{bmatrix} a \\ 2 \end{bmatrix}_q h_2 C_{a,r-2} \\ & - q^{r-1}(1-q) h_1 (C_{a,r-1} - D_{a,r-1}). \end{aligned} \tag{S3.2}$$

The exact next-index formula is

$$A_{a,r,r-1}^{(B)} = q^{(r-1)(B-1)} H_r * \frac{N_{a,r}(q^B)}{(1-q)(1-q^r)}. \tag{S3.3}$$

To derive it, we expand each source term in $z = q^b$ to subleading order,

$$\begin{bmatrix} b \\ j \end{bmatrix}_q H_j = K_j z^j \big(1 - [j]_q z^{-1} + O(z^{-2})\big).$$

In the plethysm an outer power sum p_t replaces z by z^t, so it turns the correction z^{-1} into z^{-t}; only $t = 1$ contributes to the subleading term. The identities $\partial_{p_1} H_m = [m]_q H_{m-1}$ and $\partial_{p_1}(\omega H_m) = [m]_q \omega H_{m-1}$ give the marked character (S3.1). Removing the top Gaussian basis term uses the ratio

$$\frac{[z^{r-1}] \begin{bmatrix} b-B \\ r \end{bmatrix}_q}{[z^r] \begin{bmatrix} b-B \\ r \end{bmatrix}_q} = -q^B [r]_q.$$

Thus the source restriction contributes

$$\frac{q^{(r-1)(B-1)}}{1-q^r} H_r * (W_{a,r} - q^B [r]_q D_{a,r}).$$

For the target, only occupancies 1^r and $(2, 1^{r-2})$ contribute. Their sum at the next index is

$$q^{(r-1)(B-1)} [B]_q H_r[[a]_q Y] + q^{B(r-1)} [r-1]_q \begin{bmatrix} a \\ 2 \end{bmatrix}_q H_2 H_{r-2}[[a]_q Y].$$

The recovery of Paper III, Proposition 3.2, acts on the target-source difference divided by $1 - q$. Its term $j = 1$ subtracts h_1 times the top Newton coefficient at depth $r - 1$. By (S2.2), this coefficient is $q^{B(r-1)} \mathsf{P}_{a,r-1}$, where

$$\mathsf{P}_{a,r} = H_r * \frac{C_{a,r} - D_{a,r}}{1-q}.$$

Every remaining recovery term has Gaussian degree at most $r-2$. For homogeneous f of degree $r-j$, power sums give $H_r * (h_j f) = \begin{bmatrix} r \\ j \end{bmatrix}_q H_j(H_{r-j} * f)$. Applied with $j = 1, 2$ to the target and recovery terms, this identity gives (S3.3).

S3.1.2. *The positive limit.* Apart from the monomial factor, the baseline enters (S3.3) only through $w = q^B$. Set

$$\mathcal{L}_{a,r} = \frac{N_{a,r}(0)}{(1-q)(1-q^r)}, \qquad L_{a,r} = H_r * \mathcal{L}_{a,r},$$

and define the recovery correction

$$Z_{a,r} = [a]_q[a-1]_q h_2 C_{a,r-2} - h_1 C_{a,r-1} + h_1 D_{a,r-1}.$$

Setting $w = 0$ in (S3.2) gives

$$\mathcal{L}_{a,r} = \frac{[r]_q C_{a,r} - W_{a,r}}{(1-q)(1-q^r)} + \frac{q^{r-1} Z_{a,r}}{1-q^r}. \tag{S3.4}$$

The second term of (S3.4) is positive once $Z_{a,r}$ is. In $Z_{a,r}$ only $h_1 C_{a,r-1}$ is subtracted, and a tableau injection dominates it by the first term except at the column shape 1^r, where $h_1 D_{a,r-1}$ cancels the deficit.

Lemma S3.2 (Positive recovery correction)**.** *For every $a \geq 2$ and $r \geq 2$, the character $Z_{a,r}$ is Schur-positive over $\mathbb{N}[q]$.*

Proof. With the alphabet $X_a = 1 + q + \cdots + q^{a-1}$, Cauchy's identity gives $C_{a,k} = \sum_{\mu \vdash k} s_\mu(X_a) s_\mu(Y)$, so by the Pieri rule, for a noncolumn partition ν, the coefficient of s_ν in the first two terms of $Z_{a,r}$ is

$$[a]_q[a-1]_q s_{\nu/(2)}(X_a) - s_{\nu/(1)}(X_a).$$

Delete the cell $(1,2)$ from a semistandard tableau of shape $\nu/(1)$, retaining its entry. Deletion is a weight-preserving injection into pairs of an entry from X_a and a tableau of shape $\nu/(2)$. Therefore

$$s_{\nu/(1)}(X_a) \leq_{\text{coeff}} [a]_q s_{\nu/(2)}(X_a) \leq_{\text{coeff}} [a]_q[a-1]_q s_{\nu/(2)}(X_a).$$

For the column $\nu = 1^r$, the deficit is $e_{r-1}(X_a)$. Only the occupancy 1^{r-1} of $D_{a,r-1}$ contributes to $s_{1^{r-1}}$, because a block holding $j \geq 2$ labels carries the trivial action of S_j, whereas the sign character of S_j is nontrivial. The occupancy 1^{r-1} contributes $q^{c_{r-1}} \begin{bmatrix} a \\ r-1 \end{bmatrix}_q = e_{r-1}(X_a)$ to the coefficient of $s_{1^{r-1}}$, and this contribution cancels the deficit. □

Lemma S3.2 settles the second term of (S3.4). For the first, the two terms to compare for each occupancy are $[r]_q q^{n(\lambda)} Q'_\lambda$ and $q^{e(\lambda)} V_\lambda$. Their common ungraded character is $r h_\lambda$. The target degrees are at most $2n(\lambda) + r - 1$, so the same pairing argument applies whenever $e(\lambda) - 2n(\lambda) \geq r - 1$. For length $m \geq 3$, let k be the number of ones and write the other $\ell = m - k$ parts as $2 + \delta_i$ in decreasing order. With $\Delta = \sum_i \delta_i = r - 2m + k$, (S2.6) gives

$$\begin{aligned} e(\lambda) - 2n(\lambda) &\geq 2\sum\nolimits_{i=1}^{\ell} (\ell - i)\delta_i + k\Delta - \binom{k}{2} \\ &\geq (m-1)\Delta - \binom{k}{2}. \end{aligned}$$

The last inequality follows because the two sequences $\ell - i$ and δ_i are similarly ordered. Consequently

$$e(\lambda) - 2n(\lambda) - (r-1) \geq (m-2)r - 2m(m-1) + 1 + (m-1)k - \binom{k}{2}.$$

The last two terms sum to $k(2m-1-k)/2 \geq 0$ for $0 \leq k \leq m$. It suffices that

$$r \geq \left\lceil 2m + 2 + \frac{3}{m-2} \right\rceil. \tag{S3.5}$$

For $3 \leq m \leq a$, the hypotheses $a \geq 5$ and $r \geq 2a+3$ imply (S3.5); for $m = 3,4,5,6,7$ its thresholds are $11, 12, 13, 15, 17$, which is why $a \geq 5$ is needed. At length one the two terms coincide. At length two write $\lambda = (r-k, k)$. For $2 \leq k \leq \lfloor r/2 \rfloor$, $e(\lambda) - 2n(\lambda) = k(r-k-2)$, and concavity in k reduces the inequality $k(r-k-2) \geq r-1$ to its endpoints, where it holds for $r \geq 7$. The remaining shape $(r-1,1)$ has $e(\lambda) - 2n(\lambda) = r-3$, so we compare degrees directly. For $r \geq 4$,

$$Q'_{(r-1,1)} = s_{(r-1,1)} + q s_r, \qquad V_{(r-1,1)} = ([r-1]_q + 1)(s_r + s_{(r-1,1)}).$$

The target degree lists are $2, \ldots, r+1$ for the row and $1, \ldots, r$ for the hook. In either case the source degrees are $r-1, r-1, r, \ldots, 2r-3$, which dominate both lists when $r \geq 4$. Thus the occupancy comparison holds for $a \geq 5$ and $r \geq 2a+3$. Lemma S3.2 and (S3.4) imply

$$\mathcal{L}_{a,r} \in \mathbb{N}[[q]]\{s_\nu\} \tag{S3.6}$$

in this range. The finite certificates of Section S3.2 establish (S3.6) for $a = 6$ with $2 \leq r \leq 14$ and for $a = 7$ with $2 \leq r \leq 16$, covering every depth for both inner sizes.

S3.1.3. *From the limit to a finite baseline.* The comparison with the limit works only after the internal product with H_r. At $a = B = 6$ and $r = 4$, the coefficient of $q^{14} s_4$ in the rational factor of (S3.3) is -2, although the Newton coefficient itself is positive. We therefore rewrite (S3.3) as

$$q^{-(r-1)(B-1)} A^{(B)}_{a,r,r-1} = L_{a,r} - \frac{q^B \mathsf{P}_{a,r}}{1-q}. \tag{S3.7}$$

Put $\mathsf{Q}_{a,r} = (1-q)L_{a,r}$. This function is a polynomial and satisfies $\mathsf{Q}_{a,r}(1) = \mathsf{P}_{a,r}(1)$. Indeed, for every baseline $B \geq a$ with $aB \geq 2r$, the actual Newton coefficient is a polynomial by Paper III, Equation (3.5). Its normalization in (S3.7) has no negative powers, because once the monomial factor is removed, the right side of (S3.3) has a denominator with constant term one. Multiplication by $1-q$ proves both assertions.

Theorem S2.1 gives $\mathsf{P}_{a,r} \in \mathbb{N}[q]\{s_\nu\}$ in the ranges under consideration. By (S3.6), the coefficients of $L_{a,r} = \mathsf{Q}_{a,r}/(1-q)$ are nonnegative; from degree $\deg \mathsf{Q}_{a,r}$ onward they equal $\mathsf{P}_{a,r}(1)$. If $B \geq \deg \mathsf{Q}_{a,r}$, coefficients of (S3.7) below B remain nonnegative. Every later coefficient is $\mathsf{P}_{a,r}(1)$ minus a partial sum of the nonnegative coefficients of $\mathsf{P}_{a,r}$, and is therefore nonnegative as well.

It remains to bound $\deg \mathsf{Q}_{a,r}$. The occupancy formulas give

$$\deg C_{a,r} \leq (a-1)r, \qquad \deg D_{a,r} \leq c_r + c_a, \qquad \deg W_{a,r} \leq c_r + c_a + r - 1.$$

For $D_{a,r}$, the Gaussian degree plus the outer coinvariant degrees is at most c_a; marking an occupied block adds at most $j - 1 \leq r-1$. The target and recovery terms obey the same bound, using $(a-1)r \leq c_r + c_a$, which is equivalent to $(r-a)(r-a+1) \geq 0$ for integer a, r. Thus $\deg N_{a,r}(0) \leq c_r + c_a + r - 1$. Internal multiplication by H_r multiplies every power-sum coefficient by a polynomial of degree c_r. Exact division by $1-q^r$ gives

$$\deg \mathsf{Q}_{a,r} \leq 2c_r + c_a - 1 = r(r-1) + \binom{a}{2} - 1,$$

and with the previous paragraph this bound proves Theorem S3.1. Every such B satisfies $B \geq a$ and $aB \geq ar \geq 2r$, because $a \geq 5$ and $r \geq 2$.

S3.2. **The limiting series at the next index.** Below the structural range $r \geq 2a+3$, the limit (S3.6) remains to be checked for $a=6$ with $2 \leq r \leq 14$ and for $a=7$ with $2 \leq r \leq 16$. This check is finite, because a series with denominator $1-q^r$ is nonnegative once finitely many partial sums are. We compute $C_{a,r}$, $D_{a,r}$, and $W_{a,r}$ from (S2.1) and (S3.1), form $N_{a,r}(0)$ by (S3.2), and write each polynomial $f_\nu = [s_\nu]N_{a,r}(0)/(1-q)$, for $\nu \vdash r$, as $\sum_{i=0}^{d} f_{\nu,i}q^i$. The coefficient of q^n in $f_\nu/(1-q^r)$ is

$$\sum_{\substack{0\leq i\leq \min(n,d)\\ i\equiv n \pmod r}} f_{\nu,i}.$$

For $n \geq d$, this coefficient is the full sum over the residue class of n modulo r. It is therefore enough to check these partial sums through degree d and the full sums in all r residue classes.

For $a=6$ there are 506 polynomials f_ν with 31,157 coefficients, and for $a=7$ there are 913 with 78,577 coefficients. The same computation confirms the degree bound for $\mathsf{Q}_{a,r}$ and the identity $\mathsf{Q}_{a,r}(1) = \mathsf{P}_{a,r}(1)$ of Section S3.1. Newton coefficients recovered directly from restricted tails agree with (S3.3) in 189 rational cases for $a=6$ and 123 for $a=7$, at several baselines.

Part 2. Hook coefficients and couplings

S4. Hook coefficients and matrix couplings

Hooks have unbounded depth, so the bounded-depth certificates of Paper III, Section 3 do not reach them. Specializing the alphabet to $1-u$ annihilates every other Schur function and records each hook by a power of u; this specialization gives exact rectangular coefficients and writes the one-row plethysms as positive sums over binary matrices, which we compare by pairing matrices of the same weight.

S4.1. **Rectangular hooks.** The specialization to $1-u$ factors every modified Macdonald polynomial, so it gives the first hook coefficients exactly even when the inner one-row shapes are replaced by rectangles. The modified Macdonald evaluation formula is

$$\widetilde{H}_\mu[1-u;q,t] = \prod\nolimits_{(i,j)\in\mu}(1-uq^jt^i), \tag{S4.1}$$

with zero-based coordinates; see Macdonald (1995); Haiman (2001). For homogeneous f of degree $N \geq 1$,

$$f[1-u] = (1-u)\sum\nolimits_{r=0}^{N-1}(-u)^r[s_{(N-r,1^r)}]f. \tag{S4.2}$$

Indeed, $h_j[1-u] = 1-u$ for $j \geq 1$. If $\lambda_2 \geq 2$, the first two Jacobi–Trudi rows coincide; for a hook, row elimination gives $s_{(N-r,1^r)}[1-u] = (1-u)(-u)^r$. For $N = abk$, put

$$\mathcal{F}^{(k)}_{a,b} = \frac{H_b[\widetilde{H}_{(a^k)}] - H_a[\widetilde{H}_{(b^k)}]}{1-q}.$$

Define $\vartheta_{a,b} = q^{a-1}[a]_q[b]_q[b-a]_q$, $\alpha = [a-1]_q$, and $\beta = [b-1]_q$; by Paper III, Equation (A.1), $\Gamma_{a,b} = \vartheta_{a,b}\alpha\beta/[3]_q$. At $k=1$ the inner rectangle is a row and $\mathcal{F}^{(1)}_{a,b} = F_{a,b}$. For $k \geq 2$ the hook coefficient at depth two no longer vanishes.

Proposition S4.1 (Hooks of rectangular plethysms)**.** *For $2 \leq a < b$ and $k \geq 1$, $[s_{(N)}]\mathcal{F}^{(k)}_{a,b} = [s_{(N-1,1)}]\mathcal{F}^{(k)}_{a,b} = 0$, and*

$$[s_{(N-2,1,1)}]\mathcal{F}^{(k)}_{a,b} = \vartheta_{a,b}\,t\begin{bmatrix}k\\2\end{bmatrix}_t \in \mathbb{N}[q,t]. \tag{S4.3}$$

For $k=2$ the next hook is

$$[s_{(N-3,1,1,1)}]\mathcal{F}^{(2)}_{a,b} = q\Gamma_{a,b}(1+t^3) + \vartheta_{a,b}((1+q)\alpha\beta+\alpha+\beta-1)t + \vartheta_{a,b}((1+q)\alpha\beta+\alpha+\beta)t^2 \in \mathbb{N}[q,t]. \tag{S4.4}$$

Proof. Put $K = 1+t+\cdots+t^{k-1}$ and $Z=[a]_qK$. Formula (S4.1) gives $E(u) = 1-uZ+u^2e_2[Z]-u^3e_3[Z]+O(u^4)$ for the inner rectangle. Since $H_n[X+Y]=\sum_j \begin{bmatrix} n \\ j \end{bmatrix}_q H_j[X]H_{n-j}[Y]$ and $H_j[1]=1$, we have $H_n[1+Y]=\sum_j \begin{bmatrix} n \\ j \end{bmatrix}_q H_j[Y]$, which gives

$$H_n[E(u)] = 1-[n]_qZu+S_2(n,Z)u^2-S_3(n,Z)u^3+O(u^4),$$

where, writing $d_{n,j}=\prod_{h=0}^{j-1}(1-q^{n-h})$,

$$S_2(n,Z) = [n]_qe_2[Z]+d_{n,2}e_2[Z/(1-q)],$$
$$S_3(n,Z) = [n]_qe_3[Z]+[n]_q[n-1]_qZe_2[Z]+d_{n,3}e_3[Z/(1-q)].$$

All Adams operations act on u, q, and t; thus a negative term $-uZ$ is a virtual alphabet, not the one-letter alphabet $(-u)Z$. Let $\Delta S_i = S_i(b,[a]_qK)-S_i(a,[b]_qK)$. Since $[b]_q[a]_qK=[a]_q[b]_qK$, the terms of order at most one in u cancel in the difference. By (S4.2), this cancellation gives the two vanishing coefficients, and the two desired hook coefficients are $\Delta S_2/(1-q)$ and $(\Delta S_3-\Delta S_2)/(1-q)$. After the substitution $2e_2=p_1^2-p_2$ in the former, the coefficients of $p_1[K]^2$ and $p_2[K]$ are $\vartheta_{a,b}/2$ and $-\vartheta_{a,b}/2$, respectively, so the hook coefficient is $\vartheta_{a,b}e_2[K]$, proving (S4.3). For $k=2$, substituting $K=1+t$ and $6e_3=p_1^3-3p_1p_2+2p_3$ in the latter expression, collecting $1,t,t^2,t^3$, and using $[b]_q-[a]_q=q^a[b-a]_q$ gives (S4.4). Finally $\alpha+\beta-1\in\mathbb{N}[q]$, and positivity of Γ follows from Paper III, Theorem A.1. □

S4.2. **A positive model for every hook.** Comparing the two plethysms hook by hook needs a positive formula for each hook coefficient, and (S4.2) has signs. For a one-row inner function, the product in (S4.1) expands over the subsets of $\{0,\dots,n-1\}$, and a major-index identity for words in an ordered alphabet of these subsets makes the hook evaluation positive term by term.

Let $n,m\geq 2$. For a subset $J\subseteq\{0,\dots,n-1\}$, put $w(J)=\sum_{i\in J} i$ and $\epsilon(J)=|J| \bmod 2$. Arrange these subsets in pairs $\{K, K\cup\{0\}\}$ with $K\subseteq\{1,\dots,n-1\}$. The pairs may be placed in any order, and within each pair the odd subset precedes the even subset; the result is a total order on the subsets. For a word $J_1,\dots,J_m$, define

$$\mathrm{Des}(J_1,\dots,J_m) = \{j : J_j > J_{j+1} \text{ or } J_j = J_{j+1} \text{ with } \epsilon(J_j)=1\},$$
$$\mathrm{maj}(J_1,\dots,J_m) = \sum\nolimits_{j\in\mathrm{Des}(J_1,\dots,J_m)} j.$$

The positions in this definition are numbered from 1 to m. Let $B=(B_{ij})$ be an n by m binary matrix whose columns give the subsets $J_{j+1}=\{i : B_{ij}=1\}$, with row and column indices starting at zero. The degree of B is

$$\deg_{n,m}(B) = \sum\nolimits_{i=0}^{n-1}\sum\nolimits_{j=0}^{m-1} iB_{ij} + \mathrm{maj}(J_1,\dots,J_m).$$

The double sum is the *internal degree* of B, and its *weight* is the number of ones, $\sum_{i,j}B_{ij}$. The condition $B_{00}=0$ in the next theorem accounts for the factor $1-u$ in (S4.2).

Theorem S4.2 (A positive model for every hook)**.** *For $n,m\geq 2$ and $0\leq r\leq nm-1$,*

$$[s_{(nm-r,1^r)}]H_m[H_n] = \sum_{\substack{B\in\{0,1\}^{n\times m}\\ B_{00}=0,\ \sum B_{ij}=r}} q^{\deg_{n,m}(B)}.$$

The theorem holds for every ordering of the pairs described above, although the individual statistics depend on that order. At $q = 1$, the sum counts $\binom{nm-1}{r}$ matrices, the hook dimension.

Proof. The alphabet $1-u$ carries a negative letter, so we first prove a major-index identity for words in alphabets with odd and even letters. Let $\mathcal{A}$ be a finite totally ordered alphabet with a parity $\epsilon(\alpha) \in \{0,1\}$ and an independent weight x_α on each letter. Give its power sums the superalgebra convention

$$p_\ell[\mathcal{A}] = \sum\nolimits_{\alpha\in\mathcal{A}} (-1)^{(\ell-1)\epsilon(\alpha)} x_\alpha^\ell.$$

Thus an odd letter has the generating factor $1 + zx_\alpha$ in the complete-function series, and an even letter has $(1 - zx_\alpha)^{-1}$. Then

$$H_m[\mathcal{A}] = \sum\nolimits_{(\alpha_1,\dots,\alpha_m)\in\mathcal{A}^m} q^{\mathrm{maj}(\alpha_1,\dots,\alpha_m)} x_{\alpha_1}\cdots x_{\alpha_m},$$

where descents are defined exactly as above.

Indeed, $h_m[\mathcal{A}/(1-q)]$ counts multisets of pairs (α, k) with $k \geq 0$, weighted by $x_\alpha q^k$. Repeated pairs are permitted for even letters and forbidden for odd letters. We sort these pairs by decreasing k and break ties by increasing α. The first coordinates then form a letter word, and for a fixed letter word the allowed sequences of second coordinates are

$$k_1 \geq \cdots \geq k_m \geq 0, \qquad k_j > k_{j+1} \quad \text{when } j \in \mathrm{Des}\,.$$

The strict inequalities are precisely the conditions needed to enforce the ordering at ties and exclude a repeated odd pair. With $k_{m+1} = 0$, we have $\sum_{i=1}^m k_i = \sum_{j=1}^m j(k_j - k_{j+1})$, so summing over these coordinates gives $q^{\mathrm{maj}}/(q;q)_m$. Multiplication by $(q;q)_m$ proves the word identity. It remains valid when the independent weights are specialized to monomials involving q.

The one-row case of (S4.1) gives $H_n[1-u;q] = \prod_{i=0}^{n-1}(1-uq^i)$. To apply (S4.2), we use the subset alphabet $\mathcal{A}_n(v)$ whose letter J has parity $|J| \bmod 2$ and weight $v^{|J|}q^{w(J)}$. Its power sums are

$$p_\ell[\mathcal{A}_n(v)] = \prod\nolimits_{i=0}^{n-1}(1-(-v)^\ell q^{\ell i}).$$

These products are the power sums of $H_n[1-u;q]$ after the ordinary substitution $u = -v$. The substitution is made after the plethystic Adams operation; this order accounts for the signs. By (S4.2) and the word identity,

$$(1+v)\sum\nolimits_{r=0}^{nm-1} v^r [s_{(nm-r,1^r)}] H_m[H_n] = H_m[\mathcal{A}_n(v)] = \sum\nolimits_{J_1,\dots,J_m} v^{\sum|J_j|} q^{\sum w(J_j)+\mathrm{maj}(J_1,\dots,J_m)}.$$

To cancel the factor $1+v$, we toggle the membership of 0 in the first letter J_1. This toggle is a fixed-point-free involution that preserves the power of q. Each $w(J_j)$ is unchanged because the toggled element is zero. Every descent except possibly the first is unchanged. For the first descent, if the two letters belong to different pairs, their comparison is unchanged. If they belong to the same pair, whose odd member precedes its even member, a descent occurs exactly when the second letter is odd, whatever the first.

Each involution pair consequently contributes $1+v$ times the term of its representative, the word with $0 \notin J_1$. Cancelling $1+v$ leaves exactly the binary matrices with $B_{00} = 0$, proving the theorem. □

S4.3. **A degree comparison by transposition.** Theorem S4.2 writes both hook coefficients as sums over binary matrices of the same weight, $a \times b$ matrices for the target $H_b[H_a]$ and $b \times a$ matrices for the source $H_a[H_b]$, and transposition is a bijection between the two sets. Pairing each matrix with its transpose compares the two coefficients up to a shift linear in the weight. The comparison needs every column descent to enter a nonempty column, so we order the base subsets K by decreasing binary value $\sum_{i\in K} 2^{i-1}$, still placing the odd member before the even member within each pair; the empty subset is then the largest letter.

Corollary S4.3 (A hook-dependent shift)**.** *For $a, b \geq 2$ and $0 \leq r \leq ab-1$, put $\delta = (a-1)\max\{r-1, 0\}$, $u_r(q) = [s_{(ab-r,1^r)}]H_b[H_a]$, and $v_r(q) = [s_{(ab-r,1^r)}]H_a[H_b]$. Then*

$$\frac{u_r(q) - q^\delta v_r(q)}{1-q} \in \mathbb{N}[q].$$

More precisely, with this order, matrix transposition gives the pointwise bound

$$\deg_{a,b}(B) \leq \deg_{b,a}(B^{\mathsf{T}}) + \delta.$$

Proof. The assertion is immediate when $r = 0$. Suppose $r \geq 1$. Every column descent enters a nonempty column, because the empty subset is the largest letter and is even. Therefore, writing the column indices from zero,

$$\mathrm{maj}(\text{columns of } B) \leq \sum_{j=0}^{b-1} j \sum_{i=0}^{a-1} B_{ij}.$$

The right side is the internal degree of the transposed matrix. Let i_0 be the first occupied row of B. The major index of the rows of B is at least i_0, because for $i_0 > 0$ the transition from the preceding empty row to this nonempty row is a descent at position i_0. On the other hand, one of the r ones is in row i_0, while all remaining row indices are at most $a-1$. Hence

$$\sum_{i,j} iB_{ij} \leq i_0 + (a-1)(r-1) \leq \mathrm{maj}(\text{rows of } B) + (a-1)(r-1).$$

Adding the two inequalities proves the pointwise bound. Transposition bijects the two matrix sets at fixed r, so each paired contribution to the divided difference is

$$\frac{q^{\deg_{a,b}(B)} - q^{\deg_{b,a}(B^{\mathsf{T}})+\delta}}{1-q} = q^{\deg_{a,b}(B)}[\deg_{b,a}(B^{\mathsf{T}}) + \delta - \deg_{a,b}(B)]_q \in \mathbb{N}[q].$$

Here $[0]_q = 0$. Summing proves the corollary. □

The pointwise shift is sharp for the pairing by transposition when $1 \leq r \leq b$. If the r ones fill the first r cells of the last row, the two internal degrees are $(a-1)r$ and $\binom{r}{2}$, and the two major indices are $\binom{r}{2}$ and $a-1$, respectively, so the degrees differ by $(a-1)(r-1)$.

For $a = 6$ and the hooks $(5b, 1^b)$, the corollary gives the shift $5(b-1)$, smaller than the global shift $30\lfloor b/2 \rfloor$ of Paper I (Sra, 2026a, Theorem 1.3).

S4.4. **Constraints on an unshifted coupling.** Without the shift of Corollary S4.3, the pairing argument would put every hook coefficient of $F_{a,b}$ in $\mathbb{N}[q]$. Call a bijection from the target matrices of Theorem S4.2 onto the source matrices of the same weight an *unshifted coupling* if it never lowers the degree. Two degree obstructions show that such a coupling cannot be induced by one bijection of cells, and that it cannot map every matrix into the row/column-permutation orbit of its transpose.

Proposition S4.4 (No coupling induced by a cell bijection)**.** *Suppose $n, m \geq 2$ and $n \neq m$, and order both alphabets as in Section S4.3. No bijection from the $nm-1$ cells other than the origin of an n-by-m rectangle to those of an m-by-n rectangle induces a map ϕ on subsets with*

$$\deg_{n,m}(B) \leq \deg_{m,n}(\phi B) \tag{S4.5}$$

on both the singleton and two-element layers. Thus a single inclusion-preserving bijection of the Boolean lattices cannot give all the unshifted hook comparisons in this model.

Proof. The singleton at (i, c) has degree $i + c$. For two distinct cells $(i, c), (k, d)$, with $c \leq d$, their degree is the sum of their singleton degrees minus

$$\begin{cases} c, & c = d, \\ d\,\mathbf{1}_{i>k}, & d = c+1, \\ 0, & d \geq c+2. \end{cases} \tag{S4.6}$$

Indeed, a nonempty column preceded by an empty column contributes its column index to the major index. For adjacent singleton columns the second contribution is lost exactly when the first row index is larger. A column containing two ones is even and contributes its index only once.

The sums of degrees over the singleton and pair layers are respectively

$$\frac{nm(n+m-2)}{2}, \qquad (nm-2)\frac{nm(n+m-2)}{2} - 2\binom{n}{2}\binom{m}{2}. \tag{S4.7}$$

For the second formula, the total same-column defect and the total adjacent-column defect each equal $\binom{n}{2}\binom{m}{2}$. Deleting $(0,0)$ changes neither defect sum. Both expressions in (S4.7) are symmetric in n, m. Thus (S4.5) would force equality on every singleton and pair.

The two cells whose singletons have degree one are $x = (0,1)$ and $y = (1,0)$. Set $G = [n]_q[m]_q$, and let P_c be the degree generating polynomial of the pairs containing a cell c. Equality on singletons forces the cell map to send $\{x, y\}$ onto the two corresponding cells of the transposed rectangle, and equality on pairs then makes P_c equal to the corresponding polynomial of the image of c. By (S4.6),

$$P_x = qG - q^n - q^{n+1}, \qquad P_y = qG - 2q^2.$$

For example, the baseline $qG - q - q^2$ for x changes by $\sum_{i=1}^{n-1}(q^i - q^{i+2})$; for y only its pair with x changes, by $q - q^2$. Transposing the dimensions replaces n by m in P_x and leaves P_y unchanged. The polynomial $q^n + q^{n+1}$ never equals $2q^2$ for $n \geq 2$, and it determines n. Consequently the multisets $\{P_x, P_y\}$ of the two rectangles differ when $n \neq m$, contradicting the required equality. □

Proposition S4.4 concerns a single cell map, which induces the coupling at every weight. A bijection chosen separately at each weight escapes this obstruction, but on the weight-b layer it still cannot keep every matrix in the row/column-permutation orbit of its transpose.

Proposition S4.5 (No coupling within transpose orbits)**.** *Let $2 \leq a \leq b$, and order the two alphabets in any of the permitted ways. On the weight-b layer there is no unshifted coupling that maps every target matrix into the row/column-permutation orbit of its transpose.*

Proof. Let B be a target a-by-b matrix whose ones fill one row $i \geq 1$. Its column word is the constant odd letter $\{i\}$, hence

$$\deg_{a,b}(B) = bi + \binom{b}{2} \geq b + \binom{b}{2}.$$

The row/column-permutation orbit of its transpose consists of matrices with one full source column. The origin condition excludes column zero, so the full column has positive index, and each such matrix has internal degree $\binom{b}{2}$. The column word consists of empty letters and one full letter; it has at most one descent, whose index is at most $a - 1$, independently of the alphabet order. Therefore every source matrix C in this orbit has

$$\deg_{b,a}(C) \leq \binom{b}{2} + a - 1 < \binom{b}{2} + b.$$

Thus every source matrix in this orbit has smaller degree than every target matrix with one full row of positive index. □

For $a = 6$, the weight-b layer is the hook $(5b, 1^b)$. The obstruction also rules out composing transposition with maps that preserve the multiset of column letters on each side, because the target column word is constant while the source column word has one full letter, the others empty.

S4.5. **Support bounds for dense hooks.** A hook coefficient of $F_{a,b}$ lies in $\mathbb{N}[q]$ whenever every target degree is at most every source degree, with no coupling needed. Near the full matrix, the few zero entries control both degrees. For even a, each target descent can be charged to its own zero, and for odd b, so can each missing source descent. With the order of Section S4.3, we write

$$L_m(s) = \sum_{j=\max(1,m-s)}^{m-1} j = \binom{m}{2} - \binom{\max(m-s,1)}{2}, \qquad s \geq 0,$$

for the sum of the largest $\min(s, m-1)$ descent positions, which bounds the major index of a word of length m with at most s descents.

Proposition S4.6 (Degree bounds for dense hooks)**.** *Let $a, b \geq 2$, $0 \leq s \leq N-1$, and $r = N-1-s$. With the alphabet order of Section S4.3, every target matrix B and every source matrix C of weight r satisfy the bounds*

$$\begin{aligned} \deg_{a,b}(B) &\leq bc_a + (1-\varepsilon_a)L_b(s) + \varepsilon_a c_b, \\ \deg_{b,a}(C) &\geq ac_b - (b-1)s + \varepsilon_b(c_a - L_a(s)). \end{aligned} \tag{S4.8}$$

Consequently, whenever the target upper bound in (S4.8) is at most the source lower bound, $[s_{(s+1,1^r)}]F_{a,b} \in \mathbb{N}[q]$.

Proof. There are exactly s zero cells other than the origin. If a is even, let F be the full target column and $E = F \setminus \{0\}$. Then E is odd, F is even, and $E < F <$ every other letter. A descent whose left column is outside $\{E, F\}$ can be charged to a zero in a positive row of that column. Every other descent enters an E column and can be charged to the zero in row zero of that column. These charges are distinct because the first kind lies in a column outside $\{E, F\}$, whereas the second lies in an E column; within either kind the involved columns are distinct. No charge is the origin, since a right column has positive index and a first-kind charge has positive row. Hence there are at most s descents and the major index is at most $L_b(s)$. If a is odd, the bound c_b on any major index suffices. The internal degree is at most bc_a in both cases.

The full source rectangle has internal degree $a\binom{b}{2}$. Removing each extra zero decreases it by at most $b-1$, proving the internal part of the lower bound. If b is odd, the full source column F is an odd smallest letter. Every edge entering F is a descent, including an edge from another F. Thus each nondescent can be charged to a zero in its right, nonfull column. These charges are distinct and avoid the origin. At most s of the $a-1$ potential descents can be lost; the remaining major index is at least $c_a - L_a(s)$. If b is even, the nonnegativity of the major index suffices.

Both sets of matrices have cardinality $\binom{ab-1}{r}$. Under the stated support separation, any bijection pairs a target monomial q^u with a source monomial q^v having $u \leq v$; its contribution $(q^u - q^v)/(1-q) = q^u + \cdots + q^{v-1}$ lies in $\mathbb{N}[q]$. □

For $a = 6$ and every $b \geq 7$, the criterion proves positivity of the three hook coefficients with $s = 0, 1, 2$, equivalently first-row lengths $1, 2, 3$. It suffices to check $s = 2$, since the target bound increases and the source bound decreases with s. At $b = 7$ the bounds are 116 and 120. For $b \geq 8$, even dropping the parity term, their difference is $3b^2 - 22b + 5 > 0$. More generally, once the parity term is dropped, the bound $L_b(s) \leq (b-1)s$ gives the sufficient condition $0 \leq s \leq \lfloor 3b(b-6)/(2(b-1)) \rfloor$.

These hooks have height $6b - s > 2b$, so for $b \geq 350$ they also satisfy the height condition of (Sra, 2026b, Theorem C.3); the support bounds above reach them from $b = 7$.

For $a = 7$ and $b \geq 8$, the target upper bound is $21b + c_b$. Without the nonnegative parity term, the source lower bound gives $\deg_{b,7}(C) - \deg_{7,b}(B) \geq 3b(b-8) - (b-1)s$, so the coefficient in $F_{7,b}$ of every hook $(s+1, 1^{7b-1-s})$ lies in $\mathbb{N}[q]$ whenever

$$0 \leq s \leq \left\lfloor \frac{3b(b-8)}{b-1} \right\rfloor. \tag{S4.9}$$

This range grows with b and reaches $s \le 104$ at $b = 42$, which leaves the finitely many values $b \le 41$ to the exact computation of Section S4.6.

Proposition S4.7 (Seventh-power hooks)**.** *Every hook coefficient of $F_{7,b}$ belongs to $\mathbb{N}[q]$ for $7 \le b < 42$. For every $b \ge 7$, the same conclusion holds for every hook with first part at most 105.*

Proof. The exact certificate in Section S4.6 proves the finite range $8 \le b \le 41$. On the diagonal $b = 7$, $F_{7,7} = 0$. For $b \ge 42$ and the hook $(s+1, 1^{7b-1-s})$, the function $3b(b-8)/(b-1) = 3b-21-21/(b-1)$ is increasing with floor 104 at $b = 42$, so (S4.9) covers every $s \le 104$, proving the second assertion. □

S4.6. **A certificate for the seventh-power hooks.** For $8 \le b \le 41$, Proposition S4.7 needs every hook coefficient of $F_{7,b}$, and the matrix sums of Theorem S4.2 are far too large to enumerate. Instead, we combine the hook evaluation (S4.2) with the power-sum formula of Paper III, Equation (3.2); the combination gives

$$[s_{(nm-r,1^r)}]H_m[H_n] = (-1)^r[u^r]\frac{1}{1-u}\sum\nolimits_{\rho\vdash m}\frac{\kappa_m(\rho;q)}{z_\rho}\prod\nolimits_{\ell\in\rho}(u^\ell;q^\ell)_n.$$

Here $(z;Q)_n = \prod_{i=0}^{n-1}(1-zQ^i)$, and each factor can be expanded as

$$(u^\ell;q^\ell)_n = \sum\nolimits_{j=0}^{n}(-1)^j q^{\ell\binom{j}{2}}\begin{bmatrix} n \\ j \end{bmatrix}_{q^\ell} u^{\ell j}.$$

For the source $H_7[H_b]$, the sum has only fifteen terms, independently of b. For the target, a recurrence avoids enumerating the partitions of b. Put $\mathcal{U}_{n,m}(u;q) = H_m[H_n][1-u]$, where $1-u$ is a plethystic alphabet, and set $\mathcal{U}_{n,0} = 1$. Newton's identity for h_m gives

$$m\mathcal{U}_{n,m} = \sum\nolimits_{\ell=1}^{m}\frac{(q;q)_m}{(1-q^\ell)(q;q)_{m-\ell}}(u^\ell;q^\ell)_n\,\mathcal{U}_{n,m-\ell}. \tag{S4.10}$$

Indeed, $p_\ell[H_n][1-u] = (u^\ell;q^\ell)_n$, and the outer alphabet $H_n/(1-q)$ supplies the factor $(1-q^\ell)^{-1}$. The remaining quotient is a polynomial because one of the ℓ consecutive integers $m-\ell+1,\dots,m$ is a multiple of ℓ. The hook coefficient is $(-1)^r[u^r]\mathcal{U}_{n,m}/(1-u)$.

For each $8 \le b \le 41$, we evaluate (S4.10) at $n = 7$ for the target and the sum over $\rho \vdash 7$ for the source, and we divide the difference of their hook coefficients by $1-q$ through cumulative sums. The calculation verifies exact divisibility, integral coefficients, and nonnegativity for all 5,831 hooks, examining 11,869,562 coefficient positions. It also checks each ungraded hook multiplicity against $\binom{7b-1}{r}$.

The target is computed with exact integer coefficients. For the source, the coefficients are packed into single integers in a fixed radix. Write $\|g\|_1$ for the sum of the absolute values of the coefficients of g. Since $\|\prod_{\ell\in\rho}(u^\ell;q^\ell)_n\|_1 \le 2^{n\ell(\rho)} \le 2^{nm}$, every coefficient of $m!$ times the sum over $\rho \vdash m$ has absolute value at most $2^{nm}\sum_{\rho\vdash m}(m!/z_\rho)\|\kappa_m(\rho;q)\|_1$, so a radix larger than twice this bound recovers each signed coefficient exactly. The degree bound $mc_n + c_m$ fixes the number of digits, and divisibility by $m!$ is checked coefficientwise.

As independent checks, unpacked coefficient arrays give the same hook coefficients for $8 \le b \le 26$, and direct cycle-index calculations of both sides agree at $b = 8, 9$.

Part 3. Exchange operators and graded kernels

S5. The exchange gain and degree cutoffs

The graded characters $P_{a,b}$ and $Q_{a,b}$ can be realized by row and column degree operators on the same regular module. Adding a row changes the column degree by a positive exchange gain and a loss (Section S6.8). The gain can be inverted off its kernel. The kernel is invariant under the column degree, so the inversion works inside every cutoff of that degree.

S5.1. **Positive gain and its common kernel.** Each transposition contributes a positive semidefinite summand to the gain, so the kernel of the gain is the common kernel of these summands.

Let $m \geq 3$, $a \geq 1$, and arrange am tensor positions in m rows and a columns. For a transposition $t \in S_m$, let $t^{(c)}$ act in column c. Define

$$J_t = \sum_{c=1}^{a} t^{(c)}, \qquad R_t = \prod_{c=1}^{a} t^{(c)}, \qquad \Psi_t = (a-1)I - J_t + R_t.$$

Thus R_t is the simultaneous row transposition. On a finite-dimensional unitary representation, set

$$G = \sum_t w_t \Psi_t, \qquad w_t > 0, \qquad \mathcal{K} = \ker G. \tag{S5.1}$$

All transpositions are included, so the conclusions apply to the strictly positive Hermite-weighted sums of the exchange recurrence. The weights need not be those of the degree operator B below.

For a fixed t, the involutions $t^{(c)}$ commute. On a simultaneous eigenspace with p negative signs,

$$\Psi_t = 2p - 1 + (-1)^p = 4\lfloor p/2 \rfloor. \tag{S5.2}$$

Consequently $\Psi_t \succeq 0$, and

$$\mathcal{K} = \bigcap_t \ker \Psi_t = \ker \Gamma, \qquad \Gamma := \sum_t \Psi_t.$$

The diagonal group permutes the operators Ψ_t by conjugation, so it preserves $\mathcal{K}$. On this kernel,

$$J_t = (a-1)I + R_t. \tag{S5.3}$$

In particular, each J_t preserves $\mathcal{K}$ as well.

The kernel is therefore independent of the positive weights. Theorem S6.1 in Section S6.1 classifies it on each ordered column type.

S5.2. **The column degree and compression of the gain.** A cutoff of the coinvariant grading makes sense on the regular module once that grading acts there as a self-adjoint operator commuting with the left action. For S_s we use the Hermite-weighted Polychronakos–Frahm degree operator in the coinvariant normalization,

$$L_s = \sum_{1 \leq i < j \leq s} \frac{I - (ij)}{(x_i - x_j)^2}, \qquad x_1, \ldots, x_s \text{ roots of the physicists' Hermite polynomial.}$$

Set $L_0 = L_1 = 0$. On the Specht module V_ρ, the degree character is

$$\mathrm{Tr}_{V_\rho}\, q^{L_s} = \mathfrak{f}_\rho(q) := \sum_{T \in \mathrm{SYT}(\rho)} q^{\mathrm{maj}(T)} = q^{n(\rho)} \frac{[s]_q!}{\prod_{u \in \rho} [h(u)]_q}, \quad \rho \vdash s, \tag{S5.4}$$

where $\mathfrak{f}_\varnothing = 1$. The coinvariant–Hermite intertwiner of Theorem S6.5 proves this equivariant identification. Polychronakos (1994) computed the spectrum of the corresponding spin chain.

S5.2.1. *The wreath-product grading.* The column degree must have the plethysm $Q_{a,m} = H_a[H_m]$ as its graded character, so it combines the operator L_m of each column with the operator L_a on whole columns. On the common regular module $\mathcal{H} = \mathbb{C}[S_{am}]$, with these operators acting on the right, define the degree operator

$$B = B_{\text{in}} + B_{\text{out}} = \sum_{c=1}^{a} L_m^{(c)} + L_a^{\text{out}}, \tag{S5.5}$$

where outer transpositions exchange whole columns. To justify the full plethystic grading, write $W_{m,a} = S_m \wr S_a$ and identify

$$\mathbb{C}[W_{m,a}] \cong \mathbb{C}[S_m]^{\otimes a} \otimes \mathbb{C}[S_a]$$

by writing an element as $k\sigma$. The left outer action permutes the inner factors and acts on the outer regular factor. Although individual right inner actions are twisted by σ, the sum of identical inner degree operators is invariant under this twist. Thus B_{in} is the sum of the inner degrees and B_{out} is the outer degree; in particular, they commute. We apply the intertwiner of Theorem S6.5 to the inner and outer factors and obtain

$$(\mathbb{C}[W_{m,a}], B) \cong R_m^{\otimes a} \otimes R_a$$

as graded left $W_{m,a}$-modules. Induction to S_{am} proves

$$\text{Frob}_q(\mathcal{H}, B) = H_a[H_m] = Q_{a,m}.$$

An outer cycle of length r traces the inner degree through r tensor positions, so its Adams operation sends the inner grading variable q to q^r as well as p_j to p_{rj}. The outer coefficient retains its own degree, which is why the Adams operations act on q in every character formula below.

S5.2.2. *Inverting the gain inside a cutoff.* Off its kernel the gain is bounded below. Proposition S6.3 proves $G \succeq \gamma_m w_{\min}(I - P_{\mathcal{K}})$, where $P_{\mathcal{K}}$ is the orthogonal projector onto $\mathcal{K}$, $w_{\min} = \min_t w_t$, and

$$\gamma_m = \begin{cases} 3, & m = 3, \\ 2, & m = 4, \\ 1, & m \geq 5. \end{cases}$$

The kernel reduces B_{in} and B_{out}, so this bound survives compression by the cutoffs of B.

Theorem S5.1 (No new kernel under B-compression)**.** *The kernel $\mathcal{K}$ reduces both B_{in} and B_{out}. Consequently $P_{\mathcal{K}}$ commutes with every cutoff $E_k = \mathbf{1}_{\{B \leq k\}}$. As an operator on $E_k\mathcal{H}$, let $G_k = (E_k G E_k)|_{E_k\mathcal{H}}$. Then*

$$\ker G_k = E_k\mathcal{K}, \qquad \operatorname{im} G_k = E_k(\mathcal{K}^{\perp}),$$
$$\operatorname{rank} G_k = \dim(E_k\mathcal{H}) - \dim(E_k\mathcal{K}), \qquad \|G_k^{\dagger}\| \leq (\gamma_m w_{\min})^{-1}.$$

Here $G_k^{\dagger}$ is the Moore–Penrose inverse. These statements hold separately on every left Specht multiplicity space.

Proof. The diagonal S_m preserves $\mathcal{K}$, and every outer column permutation commutes with each Ψ_t. Thus both groups preserve the kernel. Summing (S5.3) with the weights of L_m gives

$$B_{\text{in}}|_{\mathcal{K}} = L_m^{\text{diag}}\Big|_{\mathcal{K}}. \tag{S5.6}$$

Self-adjointness makes these invariant subspaces reducing, proving the commutation with E_k. For $v \in E_k\mathcal{H}$,

$$\langle v, G_k v\rangle = \langle v, Gv\rangle \geq \gamma_m w_{\min} \|(I - P_{\mathcal{K}})v\|^2.$$

Also G_k kills $E_k\mathcal{K}$, and the assertions follow. The same argument applies within each multiplicity space because all operators commute with the left S_{am}-action. □

In particular, every $y \in E_k(\mathcal{K}^\perp)$ has a unique solution $x \in E_k(\mathcal{K}^\perp)$ to $E_k G x = y$, with $\|x\| \le (\gamma_m w_{\min})^{-1}\|y\|$. The gain is thus inverted inside the prescribed B-cutoff, and the cutoff E_k itself belongs to the finite algebra generated by the exchanges (Proposition S6.6).

S5.3. **The row and column degrees.** The conjecture becomes a comparison of cutoffs once $P_{a,b}$ and $Q_{a,b}$ are realized by degree operators on one regular module. For $b \ge a \ge 2$, define the row and column degrees on the grid of b rows and a columns by

$$A_{a,b} = \sum\nolimits_{r=1}^{b} L_a^{\text{row } r} + L_b^{\text{whole rows}}, \qquad B_{a,b} = \sum\nolimits_{c=1}^{a} L_b^{\text{column } c} + L_a^{\text{whole columns}}.$$

Their graded characters are $P_{a,b}$ and $Q_{a,b}$, respectively. The same row construction on the transposed grid is denoted $A_{b,a}$; grid transposition conjugates it to $B_{a,b}$. With $m = b$, the column operator $B_{a,b}$ is the operator B in (S5.5). Interchanging rows and columns gives the same construction for $A_{a,b}$. On a Specht multiplicity space, the full conjecture is

$$\dim \mathbf{1}_{\{A_{a,b} \le k\}} S^\lambda \ge \dim \mathbf{1}_{\{B_{a,b} \le k\}} S^\lambda \qquad (\lambda \vdash ab,\ k \ge 0).$$

Theorem S5.1 controls the gain under cutoffs of $B_{a,b}$ alone. Averaged over the final cutoff (Section S7), the comparison has an explicit positive Gram formula, and even and odd cutoffs have equal totals.

S6. Graded kernels, incremental loss, and transfers

Theorem S5.1 gives the rank of the exchange gain under a B-cutoff as $\dim E_k\mathcal{H} - \dim E_k\mathcal{K}$, so the rank is explicit once the kernel $\mathcal{K}$ and its B-grading are known; that grading needs the outer column labels. The incremental loss needs, in addition, a separate old inner degree, which distinguishes it from the final column degree.

S6.1. **The kernel on each column type.** Since G lies in the group algebra of S_m^a, its kernel splits over the irreducible S_m^a-modules, the ordered column types. On each of them, Γ acts on every diagonal isotypic component by a scalar built from contents, so $\mathcal{K}$ consists of the components on which a content inequality becomes an equality; the Lie algebra generated by transpositions decides which components these are. For $\alpha \vdash m$, write V_α for its Specht module and put

$$d(\alpha) = c_m - \sum\nolimits_{(i,j)\in\alpha} (j - i).$$

The content deficit $d(\alpha)$ satisfies $0 \le d(\alpha) \le 2c_m$, with equality at the lower and upper endpoints only for (m) and (1^m), respectively. Write $\alpha_p = (m-p, 1^p)$ for the hook with leg length p. The standard representation Std_m is the action of S_m by coordinate permutation on $\{(z_1, \dots, z_m) : \sum_i z_i = 0\}$.

On $\mathcal{K}$ the inner degree is the diagonal degree (S5.6), so the inner grading of the kernel is fixed by its diagonal S_m-types. The following theorem determines these types on each ordered column type.

Theorem S6.1 (Classification of the gain kernel)**.** *Fix an ordered column type $V = V_{\alpha_1} \otimes \cdots \otimes V_{\alpha_a}$. The kernel of G is independent of the strictly positive weights. As a representation of the diagonal S_m, it is as follows.*

(1) If at most one column is nontrivial, the entire tensor product is the kernel.

(2) If at least two columns are nontrivial and every column is a hook α_{p_c}, put $P = \sum_c p_c$. For $P \le m-1$, the kernel is one copy of $V_{\alpha_P} \cong \bigwedge^P \mathrm{Std}_m$. For $P > m-1$, the kernel is zero.

(3) If exactly two columns are nontrivial and their types are conjugate nonhooks α, α', the kernel is one copy of sgn_{S_m}; the case includes two equal self-conjugate nonhooks.

(4) In every other configuration, the kernel is zero.

These cases are disjoint. Each nonzero kernel is irreducible and has multiplicity one within the fixed ordered column tensor product.

A kernel type can occur several times in an irreducible S_{am}-module; Section S6.6 counts them.

S6.1.1. *The content equality.* The proof of Theorem S6.1 begins with the eigenvalues of Γ, whose kernel is $\mathcal{K}$. Let Z_c be the transposition class sum in column c, and let Z_{diag} be the corresponding diagonal class sum, so that $\Gamma = (a-1)c_m I - \sum_c Z_c + Z_{\mathrm{diag}}$. Since a transposition class sum acts on V_α by the content sum $c_m - d(\alpha)$, Γ acts on a diagonal V_β-isotypic component of the fixed column tensor product as the scalar

$$\sum\nolimits_c d(\alpha_c) - d(\beta). \tag{S6.1}$$

Since $\Gamma \succeq 0$, a nonzero iterated Kronecker coefficient implies $d(\beta) \le \sum_c d(\alpha_c)$, and the kernel consists exactly of the equality components. In particular,

$$\mathcal{K} \neq 0 \quad \Longrightarrow \quad \sum\nolimits_c d(\alpha_c) \le 2c_m. \tag{S6.2}$$

The scalar (S6.1) does not show which equality components occur. The Lie algebra generated by transpositions decides which ones do, because its derived algebra acts on $\mathcal{K}$ in two ways, by the tensor and the diagonal actions, and these must agree.

S6.1.2. *The Lie algebra input.* Let $\mathfrak{g}_m$ be the Lie subalgebra of $\mathbb{C}[S_m]$ generated by transpositions, and put $f^\alpha = \dim V_\alpha$. By Marin's Theorem A and its representation description Marin (2007), the derived algebra $\mathfrak{g}'_m = [\mathfrak{g}_m, \mathfrak{g}_m]$ is the product of

- one $\mathfrak{sl}_{m-1}$ factor, acting on hook representations as exterior powers of its defining representation;
- for each unordered pair of nonself-conjugate nonhooks $\{\alpha, \alpha'\}$, an independent $\mathfrak{sl}_{f^\alpha}$ factor, acting on the pair as the defining representation and its dual;
- for each self-conjugate nonhook α, an orthogonal or symplectic factor acting by its defining representation on V_α.

Every nontrivial action in this list is confined to the indicated factor. The trivial and sign representations have zero derived action.

The tensor Lie action sends t to J_t, while the diagonal group action sends it to R_t. By (S5.3) these differ on $\mathcal{K}$ by a scalar, which disappears in every commutator, so the two actions of the derived Lie algebra agree on $\mathcal{K}$. Hence every diagonal type V_β in $\mathcal{K}$ must also have its prescribed $\mathfrak{g}'_m$-module structure under the tensor action.

The exclusion below bounds the number of columns of a nonself-conjugate nonhook type through the negative eigenspace of a transposition, so it needs that eigenspace and its complement to have dimension at least two.

Lemma S6.2. *If α is a nonself-conjugate nonhook, and r is the negative eigenspace dimension of a transposition on V_α, then $2 \le r \le f^\alpha - 2$.*

Proof. A zero negative eigenspace would make every transposition act trivially. If the negative eigenspace is a line, the commuting S_{m-2} acts on it either trivially or by sign. Frobenius reciprocity therefore places V_α in one of the inductions with Frobenius characters $h_{m-2}e_2$ and $e_{m-2}e_2$. Vertical Pieri leaves only

$$(m-1,1),\ (m-2,1,1),\ (2,2,1^{m-4}),\ (2,1^{m-2}),\ (1^m),$$

with nonexistent shapes omitted. Only $(2,2,1^{m-4})$ is a nonhook. For its conjugate $(m-2,2)$, the hook-length and content formulas give dimension $m(m-3)/2$ and negative eigenspace dimension $m-3$. The original shape thus has negative eigenspace dimension $(m-2)(m-3)/2$, which is one only at $m=4$, when $(2,2)$ is self-conjugate. Applying the same argument to α' proves the upper bound. For $m=3$ there are no nonhooks. □

S6.1.3. *Excluding all other configurations.* Inequality (S6.2) bounds the number of nonhook columns, after which the Lie factors and the content equality fix the diagonal type. For a nonself-conjugate nonhook α, write $f = f^\alpha$ and let r be as in Lemma S6.2. Taking traces of a transposition gives $fd(\alpha) = 2c_m r$ and $fd(\alpha') = 2c_m(f-r)$. If both types occur, then $d(\alpha) + d(\alpha') = 2c_m$, and (S6.2)

forces exactly one of each, all other columns trivial, and $\beta = (1^m)$. This sign representation occurs once, by Schur's lemma.

Otherwise choose the orientation so that $k \geq 1$ copies of α occur and none of α'. By (S6.2), $kr \leq f$, and hence $1 \leq k \leq f/2$. The finite-dimensional $\mathfrak{sl}_f$ actions integrate to the simply connected group SL_f, whose center acts on the column tensor product with exponent k modulo f. On a diagonal type in $\mathcal{K}$ the exponent is 0, 1, or -1, according as that type is trivial for this factor, defining, or dual. Since $f \geq 4$ and $1 \leq k \leq f/2$, only $k = 1$ is possible, with diagonal type $\beta = \alpha$. Equality in (S6.1) then forces every other column to be trivial.

A self-conjugate nonhook has $d(\alpha) = c_m$, so at most two can occur. With exactly one such column, its independent Lie factor forces $\beta = \alpha$, and equality leaves no other nontrivial column. With two, equality forces diagonal sign and all remaining columns trivial. Distinct self-conjugate types cannot supply a sign line, because their independent defining Lie factors would each have to have an invariant vector. Two equal types supply one sign line by Schur's lemma, which proves all nonhook assertions.

If all columns are hooks, the independent nonhook Lie factors act trivially, so the diagonal type is a hook. Since $d(\alpha_p) = mp$, equality forces $\beta = \alpha_P$ with $P = \sum_c p_c \leq m - 1$. As $\mathfrak{sl}_{m-1}$-modules under the tensor action, the exterior Pieri rule gives one copy of $\bigwedge^P \mathrm{Std}_m$ in $\bigotimes_c \bigwedge^{p_c} \mathrm{Std}_m$, so the V_{α_P}-component of $\mathcal{K}$ is at most one copy. There is no extra determinant twist on passing from GL_{m-1} to SL_{m-1}, because the total polynomial degree is fixed at $P \leq m - 1$. The adjoint of the nonzero wedge map gives this copy explicitly; its content equality places it in $\mathcal{K}$. Finally, with at most one nontrivial column every Ψ_t vanishes identically, which completes the proof of Theorem S6.1.

S6.2. **A rational projector and sharp spectral bounds.** The bound $G \succeq \gamma_m w_{\min}(I - P_{\mathcal{K}})$ used in Theorem S5.1 comes from the integral spectrum of Γ, which also gives the projector in closed form. For a positive operator, gap^+ denotes its smallest strictly positive eigenvalue. Positive-gap statements assume $a \geq 2$; for $a = 1$ the gain is zero.

Proposition S6.3 (The kernel projector and spectral bounds)**.** *Put $L = 4c_m\lfloor a/2 \rfloor$. On every finite-dimensional unitary S_m^a-representation, the orthogonal projector onto $\mathcal{K}$ is*

$$\boxed{P_{\mathcal{K}} = \prod_{j=1}^{L/\gamma_m} \left(I - \frac{\Gamma}{\gamma_m j} \right).}$$

The empty product for $a = 1$ is I. This projector belongs to the rational group algebra and is independent of the positive weights. With $w_{\min} = \min_t w_t$,

$$\boxed{G \succeq w_{\min}\Gamma \succeq \gamma_m w_{\min}(I - P_{\mathcal{K}}), \qquad \|G\| \leq 4\lfloor a/2 \rfloor \sum\nolimits_t w_t.} \tag{S6.3}$$

On the full regular S_{am}-module, for every $a \geq 2$,

$$\mathrm{gap}^+(\Gamma) = \gamma_m, \qquad \|\Gamma\| = L, \qquad \|G\| = 4\lfloor a/2 \rfloor \sum\nolimits_t w_t.$$

Proof. Equation (S6.1) makes the spectrum of Γ integral. For $m = 3$ all content deficits are divisible by three, and for $m = 4$ they are even. Equation (S5.2) gives $0 \preceq \Gamma \preceq LI$. Thus every positive eigenvalue is a multiple of γ_m in this interval, which proves the projector formula and the lower bounds; the norm bound follows by weighting the same local estimate. Explicit column types attain both endpoints. For $m = 3, 4$, two standard columns, with all other columns trivial, give the diagonal types $(2, 1)$ and $(2, 2)$, with eigenvalues three and two. For $m \geq 5$, the column types $(m - 1, 1)$ and $(m - 2, 2)$ contain $(m - 3, 2, 1)$ once, because tensoring with the standard representation is $\mathrm{Ind}\,\mathrm{Res} - I$ and $(m-2, 2)$ and $(m-3, 2, 1)$ have exactly one common predecessor, $(m-3, 2)$. Moving a box of content $m - 3$ to content -2 increases the deficit by $m - 1$, whereas the standard deficit is m, so the eigenvalue of Γ is one. Trivial columns pad these types to every $a \geq 2$, and all-sign columns attain $\|\Gamma\| = L$ and $\|G\| = 4\lfloor a/2 \rfloor \sum_t w_t$, since every Ψ_t then equals $4\lfloor a/2 \rfloor$. □

The inverse bound of Theorem S5.1 depends on the weights only through $w_{\min}$. The next theorem shows that this dependence cannot be improved by more than a constant factor.

Theorem S6.4 (The smallest weight determines the global gap up to a constant)**.** *For every strictly positive weight system, on the full regular S_{am}-module with $a \geq 2$,*

$$\boxed{\gamma_m w_{\min} \leq \operatorname{gap}^+(G) \leq \frac{2m^2}{m^2-2m+2}\, w_{\min}.} \tag{S6.4}$$

The lower bound holds on every representation.

Proof. Take two standard columns, and make all remaining columns trivial. Write $U = \{z \in \mathbb{R}^m : \sum_i z_i = 0\}$. For an edge $e = \{i, j\}$, put $u_e = (e_i - e_j)/\sqrt{2}$ and $\phi_e = u_e \otimes u_e$. The transposition on U is $I - 2u_e u_e^{\mathsf{T}}$, so on $U \otimes U$,

$$G = 4 \sum\nolimits_e w_e |\phi_e\rangle\langle\phi_e|.$$

Let $\mathcal{B}$ be the unoriented vertex-edge incidence matrix of the complete graph on m vertices, and let $D = \operatorname{diag}(w_e)$. The Gram matrix satisfies $4\,\mathrm{Gram}(\phi) = 2I + \mathcal{B}^{\mathsf{T}}\mathcal{B} =: A$. Since A is positive definite, the $M = \dim \operatorname{Sym}^2 U$ vectors ϕ_e are linearly independent and form a basis of $\operatorname{Sym}^2 U$, while G vanishes on $\bigwedge^2 U$. Its positive spectrum in this sector is exactly that of $D^{1/2} A D^{1/2}$.

As $\mathcal{B}\mathcal{B}^{\mathsf{T}} = (m-2)I + \mathbf{1}\mathbf{1}^{\mathsf{T}}$, Woodbury inversion gives

$$A^{-1} = \frac{1}{2} I - \frac{1}{2}\mathcal{B}^{\mathsf{T}}(mI + \mathbf{1}\mathbf{1}^{\mathsf{T}})^{-1}\mathcal{B}, \qquad (A^{-1})_{ee} = \frac{m^2-2m+2}{2m^2} =: \zeta_m.$$

For an edge e of minimum weight, the vector $v = D^{-1/2} A^{-1} e_e$ has quadratic form ζ_m and norm squared at least $\zeta_m^2 / w_{\min}$. Its Rayleigh quotient is at most $w_{\min}/\zeta_m$. This sector is reducing and occurs in the full regular module, proving the upper bound. The lower bound is (S6.3). □

For $m \geq 5$, the bounds in (S6.4) differ by a factor $2 + O(m^{-1})$ for every positive weight system, so no factor that grows with m can improve the lower bound. On $\operatorname{Sym}^2 U$ the upper bound is attained in the limit when one edge weight stays fixed and all others tend to infinity, because the unique bounded eigenvalue then tends to the Schur complement $w_e/(A^{-1})_{ee}$.

On column types of large total deficit the lower bound improves. Put $C_{\mathrm{col}} = a c_m I - \sum_c Z_c$. It commutes with G and $P_{\mathcal{K}}$, and $\Gamma = C_{\mathrm{col}} - (c_m I - Z_{\mathrm{diag}})$, where the parenthesized operator lies between zero and $2c_m I$. Hence

$$G \succeq w_{\min}(I - P_{\mathcal{K}}) \max\{\gamma_m I, C_{\mathrm{col}} - 2c_m I\}.$$

The maximum uses spectral functional calculus in C_{col}. On a column type with $\sum_c d(\alpha_c) > 2c_m$, the refinement improves the lower bound to $w_{\min} \max\{\gamma_m, \sum_c d(\alpha_c) - 2c_m\}$. The operator C_{col} also commutes with the degree B in (S5.5), so this refinement survives its cutoff compressions.

S6.3. **The coinvariant–Hermite identification.** The degree operator L_s of Section S5.2 has the fake-degree character (S5.4) because an explicit map conjugates it to the Euler operator of the coinvariant algebra.

Theorem S6.5 (The coinvariant–Hermite intertwiner)**.** *Let $R = \mathbb{C}[z_1, \ldots, z_s]$, $I_0 = (e_1, \ldots, e_s)$, and $E = \sum_i z_i \partial_i$. With s_{ij} interchanging z_i and z_j, define*

$$Q = \sum\nolimits_{i<j} \frac{(z_i - z_j)(\partial_i - \partial_j) - (1 - s_{ij})}{(z_i - z_j)^2}.$$

The operator Q maps R to itself and lowers degree by two. For the vector $x = (x_1, \ldots, x_s)$ of Hermite roots, with $(\tau x)_i = x_{\tau^{-1}(i)}$, the map

$$\Phi_s : R/I_0 \longrightarrow \mathbb{C}[S_s], \qquad [f] \longmapsto \left((e^{-Q/2} f)(\tau x)\right)_{\tau \in S_s}$$

is a left S_s-equivariant linear isomorphism satisfying $L_s\Phi_s = \Phi_s E$. The exponential is a finite sum on each polynomial.

Proof. In the variables $d = z_i - z_j$ and $u = (z_i + z_j)/2$, a term $a_r(u, \dots)d^r$ is sent by the corresponding summand of Q to $\big(2r - 1 + (-1)^r\big)a_r d^{r-2}$. The coefficients for $r = 0, 1$ vanish, proving polynomiality and degree lowering. The sum Q is equivariant and $[E, Q] = -2Q$.

The differential equation of the Hermite polynomial of degree s gives $\sum_{j\neq i}(x_i - x_j)^{-1} = x_i$. At any permuted root vector $y \in \mathcal{O} := S_s x$, it follows that

$$((E - Q)f)(y) = \sum\nolimits_{i<j} \frac{f(y) - f(s_{ij}y)}{(y_i - y_j)^2}. \tag{S6.5}$$

Thus $E - Q$ preserves the vanishing ideal I_x of $\mathcal{O}$. With $y = \tau x$ and $(r, t) = (\tau^{-1}i, \tau^{-1}j)$, $s_{ij}\tau x = \tau(r\,t)x$ and $(y_i - y_j)^2 = (x_r - x_t)^2$. Consequently (S6.5) is the right action of L_s with its fixed root weights; variable permutation gives the commuting left group action.

To show later that $e^{-Q/2}$ carries I_0 onto I_x, we first compute the leading homogeneous ideal of I_x. The ideal $J_x = (e_r - e_r(x) : 1 \le r \le s)$ vanishes on the $s!$ distinct points of $\mathcal{O}$. Its leading homogeneous ideal contains I_0, whose quotient has dimension $s!$ by the invariant regular sequence of degrees $1, \dots, s$. Evaluation onto $\mathbb{C}^{\mathcal{O}}$ is surjective. These upper and lower dimension bounds imply $J_x = I_x$ and $\mathrm{in}(I_x) = I_0$.

Since $[E, Q] = -2Q$, conjugation gives $E - Q = e^{-Q/2}Ee^{Q/2}$. The subspace $J = e^{Q/2}I_x$ is E-stable, hence graded: homogeneous components of any of its elements are projections by polynomials in E. The exponential preserves filtration and induces the identity on associated graded, so $\mathrm{in}(J) = \mathrm{in}(I_x) = I_0$. As J is graded, $J = I_0$ as a vector subspace. Therefore $e^{-Q/2}I_0 = I_x$, which proves the asserted quotient isomorphism and intertwining. □

To deduce (S5.4) from the theorem, it remains to compute the graded character of R/I_0. The equivariant Koszul resolution of the invariant regular sequence gives the graded trace of Paper III, Equation (3.2), on $R/I_0 = R_s$, so its Frobenius characteristic is $H_s = (q;q)_s h_s[X/(1-q)]$. Cauchy's identity and principal specialization give its Schur coefficient $(q;q)_s s_\rho[1/(1-q)]$, the hook formula in (S5.4); the standard-tableau q-hook formula gives the major-index expression. The left/right regular decomposition then proves (S5.4) on each individual right Specht space, with trivial degree zero and sign degree $\binom{s}{2}$.

S6.4. **The cutoffs as polynomials in the exchanges.** Theorem S5.1 compresses G by the spectral projections E_k of B. Since B is an affine combination of exchanges with integral spectrum, these projections lie in the algebra the exchanges generate, and that algebra preserves $\mathcal{K}$.

Proposition S6.6 (Cutoffs in the exchange algebra)**.** *Let $\mathcal{A}_G$ be the unital algebra generated by all J_t and whole-column transpositions. Every B-cutoff E_k belongs to $\mathcal{A}_G$, and is a linear combination of words of length at most $D = a\binom{m}{2} + \binom{a}{2}$ in these generators. The algebra preserves $\mathcal{K}$ and $\mathcal{K}^\perp$.*

Proof. Each J_t preserves $\mathcal{K}$ by (S5.3), and whole-column transpositions preserve $\mathcal{K}$ because they commute with every Ψ_t. Since these generators are self-adjoint, they also preserve $\mathcal{K}^\perp$. The operator B is an affine linear combination of these generators. Its spectrum lies in $\{0, \dots, D\}$ by the grading identification of Section S5.2. Thus

$$E_k = \sum\nolimits_{d=0}^{\min(k,D)} \prod_{\substack{0\le j\le D\\ j\neq d}} \frac{B - jI}{d - j}$$

for integral $k \ge 0$, with $E_k = 0$ for $k < 0$. Expanding this degree-D polynomial gives the word-length bound. □

S6.5. **The full B-grading on every kernel component.** The rank in Theorem S5.1 needs $\dim E_k\mathcal{K}$, that is, the B-graded character of the kernel. On $\mathcal{K}$ the inner degree is the diagonal one by (S5.6), so this character is determined by how the outer S_a permutes columns of equal type, which the wreath labels below record.

S6.5.1. *Wreath labels and outer induction.* Put $W = S_m \wr S_a$. Label its irreducibles $W_{\boldsymbol{\eta}}$ by families $\boldsymbol{\eta} = (\eta_\alpha)_{\alpha\vdash m}$ with $n_\alpha = |\eta_\alpha|$ and $\sum_\alpha n_\alpha = a$, where

$$W_{\boldsymbol{\eta}} = \mathrm{Ind}_{\prod_\alpha (S_m^{n_\alpha}\rtimes S_{n_\alpha})}^{W} \bigotimes_\alpha \left(V_\alpha^{\otimes n_\alpha} \otimes V_{\eta_\alpha}\right).$$

Permutations of columns of equal type act by ordinary tensor permutation on the first factor. Empty labels are omitted. Define

$$F_{\boldsymbol{\eta}}(q) = \mathrm{Tr}_{\ker(G|_{W_{\boldsymbol{\eta}}})}\, q^B.$$

The trace is well defined by Theorem S5.1. On each kernel $\ker(G|_{W_{\boldsymbol{\eta}}})$ the outer S_a acts through a module induced from a Young subgroup, and the outer induction identity gives the degree character of such a module. Let $\gamma_j \vdash n_j$ be partitions with $\sum_j n_j = a$, and let the bracket denote the Gaussian multinomial $[a]_q!/\prod_j [n_j]_q!$. Then

$$\mathrm{Tr}_{\mathrm{Ind}_{\prod_j S_{n_j}}^{S_a}(\boxtimes_j V_{\gamma_j})}\, q^{L_a} = \left\langle H_a, \prod_j s_{\gamma_j}\right\rangle = \begin{bmatrix} a \\ n_1,\dots,n_r\end{bmatrix}_q \prod_j \mathfrak{f}_{\gamma_j}(q). \tag{S6.6}$$

Indeed, the multidegree $(n_1,\dots,n_r)$ component of $H_a[X_1+\cdots+X_r;q]$ is

$$\frac{(q;q)_a}{\prod_j (q;q)_{n_j}} \prod_j H_{n_j}[X_j;q].$$

Taking scalar products proves (S6.6).

Theorem S6.1, (S5.6), and (S6.6) determine every polynomial $F_{\boldsymbol{\eta}}$. The only new input is the sign by which an outer transposition acts on the kernel when it exchanges two columns of equal type.

Theorem S6.7 (The B-graded characters of the kernel)**.** *The following four families exhaust the nonzero polynomials $F_{\boldsymbol{\eta}}$.*

(H) Hook columns. *Write $n_p = |\eta_{\alpha_p}|$, $P = \sum_p p n_p$, and*

$$\widehat{\eta}_p = \begin{cases} \eta_{\alpha_p}, & p \text{ even},\\ \eta'_{\alpha_p}, & p \text{ odd}.\end{cases}$$

For $P \le m-1$,

$$F_{\boldsymbol{\eta}}(q) = q^{P(P+1)/2} \begin{bmatrix} m-1 \\ P\end{bmatrix}_q \begin{bmatrix} a \\ (n_p)_p\end{bmatrix}_q \prod_p \mathfrak{f}_{\widehat{\eta}_p}(q).$$

For $P > m-1$ it is zero. Family (H) includes the all-trivial columns.

(S) One nonhook column. *For a nonhook α, take $\eta_\alpha = (1)$ and let $\eta_0 := \eta_{(m)} \vdash a-1$; all other labels are empty. Then*

$$F_{\boldsymbol{\eta}}(q) = \mathfrak{f}_\alpha(q)[a]_q \mathfrak{f}_{\eta_0}(q).$$

(D) Two distinct conjugate nonhooks. *For nonhooks $\alpha \ne \alpha'$, the labels $\eta_\alpha = \eta_{\alpha'} = (1)$ and $\eta_0 \vdash a-2$ give $F_{\boldsymbol{\eta}}(q) = q^{c_m}[a]_q[a-1]_q \mathfrak{f}_{\eta_0}(q)$.*

(E) Two equal self-conjugate nonhooks. *For $\alpha = \alpha'$ nonhook, take $\eta_\alpha \vdash 2$ and $\eta_0 \vdash a-2$. If $r(\alpha)$ is the number of diagonal boxes, put*

$$\epsilon_\alpha = (-1)^{(m-r(\alpha))/2}, \qquad \widehat{\eta}_\alpha = \begin{cases} \eta_\alpha, & \epsilon_\alpha = 1,\\ \eta'_\alpha, & \epsilon_\alpha = -1.\end{cases}$$

Then

$$F_{\boldsymbol{\eta}}(q) = q^{c_m} \begin{bmatrix} a \\ 2 \end{bmatrix}_q \mathfrak{f}_{\eta_0}(q)\mathfrak{f}_{\widehat{\eta_\alpha}}(q).$$

In every other family, $F_{\boldsymbol{\eta}} = 0$.

Proof. For an ordered hook tuple, the kernel copy is the adjoint image of the wedge map $\bigotimes_c \bigwedge^{p_c} \mathrm{Std}_m \to \bigwedge^P \mathrm{Std}_m$. Exchanging two equal p-factors changes this map by $(-1)^{p^2} = (-1)^p$. Thus, as a module for the diagonal S_m and the outer S_a, the kernel in $W_{\boldsymbol{\eta}}$ is

$$V_{\alpha_P} \otimes \mathrm{Ind}_{\prod_p S_{n_p}}^{S_a} \left(\boxtimes_p V_{\widehat{\eta_p}} \right).$$

Family (H) follows by combining (S5.6), (S6.6), and $\mathfrak{f}_{\alpha_P} = q^{P(P+1)/2} \left[{m-1 \atop P} \right]_q$.

For a single nonhook, the corresponding module is $V_\alpha \otimes \mathrm{Ind}_{S_{a-1}\times S_1}^{S_a}(V_{\eta_0} \boxtimes \mathbf{1})$, proving (S). For distinct conjugate nonhooks it is $\mathrm{sgn}_{S_m} \otimes \mathrm{Ind}_{S_{a-2}\times S_1 \times S_1}^{S_a}(V_{\eta_0} \boxtimes \mathbf{1} \boxtimes \mathbf{1})$, proving (D). No sign twist arises, because no outer transposition exchanges a column of type α with one of type α'.

For equal self-conjugate nonhooks, the ordinary tensor flip acts on the unique diagonal sign line in $V_\alpha \otimes V_\alpha$ by ϵ_α, as Marin (2007, Section 4.2, Lemmas 5–7) shows in the language of bilinear forms; we verify it directly. In a Young orthogonal basis, let $u(T)$ be the sign of the row-reading permutation of T, and define $Je_T = u(T)e_{T'}$. Adjacent transpositions change $u(T)$ by a sign while tableau transpose negates the axial distance. The Young orthogonal matrices give $J\rho(t) = -\rho(t)J$. The same identity is immediate when the two entries share a row or column. Transposing the diagram exchanges $(m - r(\alpha))/2$ pairs of boxes, so $u(T') = \epsilon_\alpha u(T)$ and $J^{\mathsf{T}} = \epsilon_\alpha J$. The vector $\sum_T e_T \otimes Je_T$ transforms by diagonal sign and has flip eigenvalue ϵ_α. Therefore the kernel in $W_{\boldsymbol{\eta}}$ is

$$\mathrm{sgn}_{S_m} \otimes \mathrm{Ind}_{S_{a-2}\times S_2}^{S_a}(V_{\eta_0} \boxtimes V_{\widehat{\eta_\alpha}}).$$

Equation (S6.6) proves (E). The classification excludes every unlisted family. □

A family contributes to $\dim E_k\mathcal{K}$ only from its least B-degree on. Every fake degree $\mathfrak{f}_\rho$ has lowest term $q^{n(\rho)}$, and every Gaussian multinomial has constant term one, so the least B-degrees in (H), (S), (D), and (E) are respectively

$$\frac{P(P+1)}{2} + \sum\nolimits_p n(\widehat{\eta}_p), \qquad n(\alpha) + n(\eta_0),$$
$$c_m + n(\eta_0), \qquad c_m + n(\eta_0) + n(\widehat{\eta}_\alpha).$$

In particular, families (D) and (E) vanish below degree c_m, and family (H) with total leg P vanishes below degree $P(P+1)/2$. The outer labels can raise these thresholds further.

S6.6. **Ranks of the compressed gain.** Summing Theorem S6.7 over the wreath labels gives $\dim E_k\mathcal{K}$ on every Specht multiplicity space, and with it the rank in Theorem S5.1 in closed form. The graded Frobenius characteristic of the kernel in the common regular S_{am}-module is

$$\boxed{\mathcal{Z}_{m,a}(X;q) = \sum\nolimits_{\boldsymbol{\eta}} F_{\boldsymbol{\eta}}(q) \prod\nolimits_{\alpha\vdash m} s_{\eta_\alpha}[s_\alpha].} \tag{S6.7}$$

By Frobenius reciprocity and wreath induction, the restriction multiplicity of $W_{\boldsymbol{\eta}}$ in S^λ is

$$m_{\lambda,\boldsymbol{\eta}} = \left\langle s_\lambda, \prod\nolimits_{\alpha\vdash m} s_{\eta_\alpha}[s_\alpha] \right\rangle.$$

The operators act trivially on these multiplicity factors, whence $\mathrm{Tr}_{S^\lambda}(P_{\mathcal{K}} q^B) = \sum_{\boldsymbol{\eta}} m_{\lambda,\boldsymbol{\eta}} F_{\boldsymbol{\eta}}(q)$, which proves (S6.7).

Corollary S6.8 (The rank of G_k)**.** *On the S^λ multiplicity space of the regular S_{am}-module, with G_k as in Theorem S5.1,*

$$\boxed{\operatorname{rank} G_k = \sum_{j=0}^{k} [s_\lambda q^j]\,(H_a[H_m] - \mathcal{Z}_{m,a}(X;q))\,.} \tag{S6.8}$$

The inverse bound of Theorem S5.1 holds on its image. For the whole left S^λ-isotypic space, multiply this rank by d_λ.

Proof. On the multiplicity space, the first sum is $\dim E_k\mathcal{H}$ and the second is $\dim E_k\mathcal{K}$; Theorem S5.1 gives the rank as their difference. □

S6.6.1. *The kernel graded by inner degree.* When only the inner degree B_{in} is graded, the wreath labels are not needed, and Theorem S6.1 and (S5.6) give the kernel character directly. For comparison, put $C_m(z) = \sum_{p=0}^{m-1} s_{\alpha_p} z^p$. The Frobenius characteristic of $\mathcal{K}$ for the inner grading is

$$\begin{aligned}\operatorname{Frob}_{q,\text{in}} \mathcal{K} = {} & \sum_{P=0}^{m-1} \mathfrak{f}_{\alpha_P}(q)[z^P]C_m(z)^a \\ & + a h_m^{a-1} \sum_{\alpha \text{ nonhook}} \mathfrak{f}_\alpha(q) s_\alpha \\ & + \binom{a}{2} q^{c_m} h_m^{a-2} \sum_{\alpha \text{ nonhook}} s_\alpha s_{\alpha'}.\end{aligned} \tag{S6.9}$$

For $a = 1$, omit the last term. Peter–Weyl decomposition on S_m^a assigns the left factor $\prod_c s_{\alpha_c}$ to an ordered column type, and its right kernel contributes $\mathfrak{f}_\beta(q)$ by (S5.6). Summing the classification proves (S6.9). The sum over all nonhooks counts both orientations of a distinct conjugate pair and counts a self-conjugate pair once. Setting $q = 1$ replaces each hook fake degree by $\binom{m-1}{P}$ and each other fake degree by f^α. The rank formula (S6.8) needs the full B-grading.

S6.7. **Sharp support of the graded gain kernel.** Inverting the gain gives no information on its kernel, so the kernel must be compared directly. For $m, a \geq 3$ the kernel types come from hook columns of total leg at most $m-1$ or from column types that include a trivial column, so the character $\mathcal{Z}_{m,a}$ of (S6.7) is supported on shapes with a long first row and bounded length. Section S6.8.1 uses this support after the graded lift cancels the trivial diagonal type for $a = 6$. From now on $m, a \geq 3$.

Theorem S6.9 (Sharp kernel support)**.** *If $[s_\lambda q^j]\mathcal{Z}_{m,a}(X;q) \neq 0$, then*

$$\boxed{\lambda_1 \geq m - \left\lfloor \frac{m-1}{a} \right\rfloor, \qquad \ell(\lambda) \leq m + a - 1, \qquad 0 \leq j \leq c_m + c_a.}$$

One partition attains both shape bounds, and the degree bound is attained.

Proof. First consider hook column types $(m - p_c, 1^{p_c})$, where $\sum_c p_c \leq m-1$. At least one p_c is at most $\lfloor (m-1)/a \rfloor$. Every constituent of the product of the column Schur functions contains each column partition by the Littlewood–Richardson rule, and hence has first row at least $\max_c(m - p_c)$. Its length is at most the sum of the column lengths, which is $a + \sum_c p_c \leq m + a - 1$. The same conclusions hold after outer symmetrization, because $s_\eta[s_\alpha]$ is a constituent of $s_\alpha^{|\eta|}$, so symmetrization only removes constituents of the tensor product.

If the only nontrivial column is a nonhook α, the other $a-1$ columns are trivial. They force first row at least m, and the length is at most $\ell(\alpha) + a - 1 \leq m + a - 3$, since a nonhook partition of m has length at most $m-2$. For a conjugate nonhook pair α, α', the assumption $a \geq 3$ leaves a trivial column, again forcing first row at least m. Since a nonhook has a cell outside its first row and column, $\alpha_1 + \ell(\alpha) \leq m$. Therefore the total length is at most $m + a - 2$ in this family. These observations prove the shape bounds.

For the degree bound, the kernel grading is the sum of the commuting diagonal S_m and outer S_a coinvariant degrees. Their ranges lie in $[0, c_m]$ and $[0, c_a]$, respectively.

To prove shape sharpness, write $m-1 = at+r$ with $0 \le r < a$. Choose $a-r$ hook columns with leg t and r with leg $t+1$. Their total leg is $m-1$, so the diagonal kernel type is sign. The union of the row multisets of the columns is

$$\lambda^* = ((m-t)^{a-r}, (m-t-1)^r, 1^{m-1}). \tag{S6.10}$$

Its Littlewood–Richardson coefficient in the column product is one, because conjugating the partitions turns this union into their componentwise sum, the Cartan constituent. At $q=1$, summing over all outer labels for this hook multiset gives

$$\frac{a!}{\prod_p n_p!} \prod_p s_{(m-p,1^p)}^{n_p},$$

where n_p is the number of leg-p columns, by $\sum_{\eta \vdash n} f^\eta s_\eta[s_\alpha] = s_\alpha^n$ and $f^{\eta'} = f^\eta$. Thus s_{λ^*} occurs in $\mathcal{Z}_{m,a}$, and λ^* attains both shape bounds.

Finally, to attain the degree bound, choose hook legs $(m-1, 0, \dots, 0)$, whose diagonal type is the sign, of degree c_m. On the $a-1$ trivial columns choose the outer stabilizer label sgn_{a-1}. Its induction to S_a contains sgn_a, of degree c_a, for a total degree $c_m + c_a$. □

The argument for the first-row bound needs $a \ge 3$, because with only two columns the conjugate nonhook family leaves no trivial column.

At the boundary shape λ^* the whole graded kernel coefficient is explicit, because the two extreme shape bounds leave only the balanced hook multiset of the sharpness proof.

Proposition S6.10 (Exact coefficient at the sharp boundary)**.** *With $m-1 = at+r$ where $0 \le r < a$, and with λ^* as in (S6.10), let*

$$s = \begin{cases} M_{a-r}, & m+t \text{ odd}, \\ M_r, & m+t \text{ even}. \end{cases}$$

Then the entire kernel coefficient is

$$[s_{\lambda^*}]\mathcal{Z}_{m,a}(X;q) = q^{c_m+s} \begin{bmatrix} a \\ r \end{bmatrix}_q. \tag{S6.11}$$

Proof. Maximal length $m+a-1$ excludes every nonhook family and forces total hook leg $m-1$. Put $L = m-t$. First-row containment forces each leg to have the form $p_i = t + d_i$, with $d_i \ge 0$ and $\sum_i d_i = r < a$. The conjugate of λ^* is $\lambda^{*\prime} = (m+a-1, a^{L-2}, a-r)$. For the highest weight $\sigma = \sum_i (m-p_i, 1^{p_i})'$ of the conjugate tensor product, $\sigma_1 = m+a-1$ and $\sigma_j \le a$ for $j \ge 2$. If some $d_i \ge 2$, then $L-1 \ge 2$ and $\sigma_{L-1} = \#\{i : d_i \le 1\} < a$. Thus σ does not dominate $\lambda^{*\prime}$, so that tensor product cannot contain the required constituent. The balanced multiset, with all d_i zero or one, is the only possibility. Its conjugate is the Cartan constituent and has multiplicity one.

On n_p identical leg-p columns, the constituent of $s_{\alpha_p}^{n_p}$ formed by the union of rows has trivial outer label if m is even and sign outer label if m is odd. Indeed, after conjugation the Cartan constituent is symmetric in identical factors, and

$$\omega(s_\eta[s_\alpha]) = \begin{cases} s_\eta[s_{\alpha'}], & |\alpha| \text{ even}, \\ s_{\eta'}[s_{\alpha'}], & |\alpha| \text{ odd}. \end{cases}$$

The twist $\widehat{\eta}_p$ of Theorem S6.7 conjugates this label when p is odd. Its fake degree is therefore $q^{M_{n_p}}$ if $m+p$ is odd, and one otherwise. Since $a-r$ columns have leg t and r have leg $t+1$, the two fake degrees multiply to q^s. Outer induction supplies $\left[\begin{smallmatrix} a \\ r \end{smallmatrix}\right]_q$, and the diagonal sign contributes q^{c_m}. These factors prove (S6.11). □

The cancellation in Section S6.8.1 concerns the case $a = 6$, $m = b$, for which Theorem S6.9 takes the following form.

Corollary S6.11 (Six-column form)**.** *For $a = 6$ and $m = b \geq 6$, the gain kernel is supported only where*

$$\lambda_1 \geq \left\lceil \frac{5b+1}{6} \right\rceil, \qquad \ell(\lambda) \leq b+5, \qquad j \leq \binom{b}{2} + 15.$$

Outside either shape bound the gain is nonsingular on the entire Specht multiplicity space. Above the degree bound, it has no kernel on the corresponding B-homogeneous subspace.

S6.8. **The exchange recurrence and the remaining comparison.** The full conjecture compares cutoffs of the row and column degrees $A_{a,b}$ and $B_{a,b}$ of Section S5.3. Let S transpose the leading $a \times a$ square and fix the other cells, and set $\mathcal{D}_b = A_{a,b}S - SB_{a,b}$. At $b = a$ the map S is the grid transposition, which conjugates $A_{a,a}$ to $B_{a,a}$, so $\mathcal{D}_a = 0$. We follow $\mathcal{D}_b$ as rows are added, because each added row changes it by a gain with positive weights, a loss and a commutator. Embed the old operators in the grid with $b+1$ rows, whose $N = a(b+1)$ cells form b old rows and one new row, and put

$$A_{\mathrm{pre}} = A_{a,b} + L_a^{\mathrm{new}}, \quad B_{\mathrm{pre}} = B_{a,b} + L_a^{\mathrm{new}}, \quad \Gamma_{\mathrm{row}} = L_{b+1}^{\text{whole rows}} - L_b^{\text{whole rows}}.$$

For a transposition $t = (c\,d)$ in S_a, let U_t exchange columns c, d in all old rows and V_t exchange their two cells in the new row. The exchanges U_t and V_t define two commuting S_a actions, and the incremental loss is

$$\delta = \sum\nolimits_t w_a(t)(I - U_t)(I - V_t) = \sum\nolimits_t w_a(t)(I - U_t - V_t + U_tV_t), \tag{S6.12}$$

where $w_a(c,d) = (x_c - x_d)^{-2}$ for the a-root Hermite system. The loss is the two-factor instance of (S5.1), with underlying group S_a. Label the Hermite roots increasingly at each size, write $w_s(i,j) = (x_i^{(s)} - x_j^{(s)})^{-2}$, and extend w_b by zero on pairs involving the new row. Define $\Delta w_{ij} = w_{b+1}(i,j) - w_b(i,j)$ and $G_\Delta = \sum_{i<j} \Delta w_{ij} \Psi_{(i\,j)}$. Direct expansion of (S5.1) and (S6.12) gives the algebraic identities

$$A_{a,b+1} = A_{\mathrm{pre}} + \Gamma_{\mathrm{row}}, \qquad B_{a,b+1} = B_{\mathrm{pre}} + \Gamma_{\mathrm{row}} + G_\Delta - \delta.$$

Since S fixes the new row, they imply

$$\mathcal{D}_{b+1} = \mathcal{D}_b + S(\delta - G_\Delta) + [\Gamma_{\mathrm{row}}, S]. \tag{S6.13}$$

With this indexing, every coefficient of G_Δ is strictly positive; the proof compares the Stieltjes identities of two consecutive root systems.

Lemma S6.12 (Positive increments of the Hermite weights)**.** *Let $x_1^{(n)} < \cdots < x_n^{(n)}$ be the zeros of the physicists' Hermite polynomial, and retain the first n labels when passing to size $n+1$. For $n \geq 2$ and $1 \leq i < j \leq n$,*

$$0 < x_j^{(n+1)} - x_i^{(n+1)} < x_j^{(n)} - x_i^{(n)}.$$

Consequently, extending w_n by zero on pairs involving the new label $n+1$, one has

$$w_{n+1}(i,j) - w_n(i,j) > 0 \qquad (1 \leq i < j \leq n+1).$$

The last assertion also holds for $n = 1$.

Proof. Assume first that $n \geq 2$. Write $x_i = x_i^{(n+1)}$, $z = x_{n+1}^{(n+1)}$, and $y_i = x_i^{(n)}$ for $1 \leq i \leq n$. The Hermite equation gives the Stieltjes identities

$$y_i = \sum\nolimits_{j \neq i} \frac{1}{y_i - y_j}, \qquad x_i = \sum_{\substack{j \neq i \\ j \leq n}} \frac{1}{x_i - x_j} + \frac{1}{x_i - z}.$$

Put $d_i = y_i - x_i$, $v_{ij} = ((x_i - x_j)(y_i - y_j))^{-1} > 0$, and let $\mathcal{L}$ be the weighted Laplacian with weights v_{ij}. Subtracting the identities gives

$$(I + \mathcal{L})d = f, \qquad f_i = \frac{1}{z - x_i}.$$

The vector f is strictly increasing, and the inequality of the lemma says exactly that d is strictly increasing as well, so we pass to differences.

Let $(\nabla u)_i = u_{i+1} - u_i$. Since $\mathcal{L}\mathbf{1} = 0$, there is a matrix K with $\nabla\mathcal{L} = K\nabla$. Expanding $u_j = u_1 + \sum_{k<j}(\nabla u)_k$ gives, off the diagonal,

$$K_{ik} = \begin{cases} \sum_{j \le k}(v_{i+1,j} - v_{i,j}), & k < i, \\ \sum_{j > k}(v_{i,j} - v_{i+1,j}), & k > i. \end{cases}$$

Every displayed summand is strictly negative, because both root lists are increasing and the weight decreases when either endpoint moves away from the other. On $\mathbf{1}^{\perp}$, the difference map ∇ is an isomorphism, and $\mathcal{L}$ is positive definite. Thus K is similar to a positive-definite matrix, and $C = I + K$ has positive eigenvalues and nonpositive off-diagonal entries.

The inverse C^{-1} is entrywise nonnegative with positive diagonal. Indeed, choose $\eta \ge \max_i C_{ii}$. The positive spectrum of C ensures convergence in

$$C^{-1} = \int_0^{\infty} e^{-tC}\, dt = \int_0^{\infty} e^{-\eta t} e^{t(\eta I - C)}\, dt,$$

and the last integrand has both properties, because the matrix $\eta I - C$ is entrywise nonnegative. It follows that $\nabla d = C^{-1}\nabla f$ is entrywise strictly positive. Summing its entries from i to $j - 1$ proves the asserted gap contraction. Reciprocal squares therefore increase on every old pair; each new pair has positive weight and zero old weight. For $n = 1$, there is only a new pair, so the conclusion is immediate. □

The labeling matters. If the middle root of the cubic Hermite polynomial were assigned to the new row, the two retained roots would have weight $1/6$, compared with $1/2$ for the quadratic one; the increment would be $-1/3$.

By Lemma S6.12, G_{Δ} is a gain operator with positive weights, and Theorem S5.1 and Corollary S6.8 give its kernel and its ranks under cutoffs of $B_{a,b+1}$. The loss is treated in Subsection S6.9. It remains to control the accumulated defect $\mathcal{D}_b$ and the commutator in (S6.13) under both the A- and the B-cutoffs; by (S6.16), isolated increments at a common cutoff do not suffice.

S6.8.1. *The ground degree and a cancellation by the graded lift.* Two parts of the gain kernel need arguments other than inverting the gain: the ground degree, and for $a = 6$ the component of trivial diagonal type. The full kernel formula gives

$$\mathcal{Z}_{m,a}(X; 0) = h_a[h_m].$$

Only all-trivial columns with trivial outer label have degree zero. Thus the entire Q ground space lies in the gain kernel, so inverting the gain gives no information on the ground spaces; their comparison is the classical one, since the constant coefficient of $F_{a,b}$ is $h_b[h_a] - h_a[h_b]$ (Bergeron, 2017).

For $a = 6$ and $b \ge 25$, the graded lift permits an exact cancellation (Paper III, Corollary 2.2). In the $Q_{6,b}$ gain kernel, the trivial diagonal S_b type forces every inner column type to be trivial, so the component of this type has full characteristic $H_6[h_b]$. The part of $P_{6,b}$ with trivial outer type has characteristic $h_b[H_6]$. The graded lift therefore places an isomorphic copy of this entire Q-kernel component inside P, degree by degree. By complete reducibility, removing $H_6[h_b]$ from both sides leaves the graded characters

$$P_{6,b} - H_6[h_b] \quad \text{and} \quad Q_{6,b} - H_6[h_b],$$

whose difference is still $P_{6,b} - Q_{6,b}$. The first difference is a graded character by (1.3) and the positive outer remainder $H_b[H_6] - h_b[H_6]$, and the second because $h_b \subset H_b$.

After this cancellation, the nontrivial diagonal kernel types and the image of the gain remain; Theorem S6.9 locates the former. Bergeron conjectured more generally that the ordinary inequality implies the graded one (Bergeron, 2017, Conjecture 10).

S6.9. **The incremental loss and its complete grading.** Each added row subtracts the loss δ from the column degree $B_{a,b+1}$, so the comparison needs its kernel and cutoff ranks as well as those of the gain. Throughout this subsection $a \geq 3$ and $b \geq 1$; the formulas below also hold for $a = 1, 2$ by direct evaluation of their trivial and sign factors. Define the old inner degree, the pair degree, the intermediate degree, and its cutoffs by

$$I_{\mathrm{old}} = \sum_{c=1}^{a} L_b^{(c)}, \qquad D_{\mathrm{pair}} = L_a^U + L_a^V, \qquad C = I_{\mathrm{old}} + D_{\mathrm{pair}}, \qquad F_k = \mathbf{1}_{\{C \leq k\}}, \tag{S6.14}$$

so that $C = B_{\mathrm{pre}}$. The old inner degree commutes with both S_a actions, and $D_{\mathrm{pair}}|_{\ker \delta} = L_a^{UV}|_{\ker \delta}$, where the diagonal degree L_a^{UV} preserves $\ker \delta = \bigcap_t \ker(I - U_t)(I - V_t)$ because the diagonal S_a permutes these factors. Thus $\ker \delta$ reduces I_{old} and D_{pair} separately, and therefore reduces C.

Write $\mathcal{E}_a$ for the set of triples (α, β, γ) of partitions of a with Kronecker coefficient $g_{\alpha\beta\gamma} > 0$ and $d_a(\gamma) = d_a(\alpha) + d_a(\beta)$, where d_a is the content deficit of Subsection S6.1 for partitions of a. Applied to two factors, Theorem S6.1 says that every such Kronecker multiplicity is one. If one input is trivial, the output is the other input; two hooks of leg lengths p, r give the hook of leg $p + r$ when $p + r \leq a - 1$; conjugate nonhooks give sign. There are no other triples. Since the loss is the two-factor instance of the gain, these triples index its kernel.

Theorem S6.13 (Bigraded kernel and cutoff ranks of the loss)**.** *Grade* $\ker \delta$ *by* I_{old} *with variable* u *and by* D_{pair} *with variable* q*. Its full left characteristic is*

$$\boxed{\mathcal{Z}_\delta(X; u, q) = \sum_{(\alpha,\beta,\gamma) \in \mathcal{E}_a} \mathfrak{f}_\gamma(q)\, s_\alpha[H_b(X; u)]\, s_\beta(X).} \tag{S6.15}$$

Set $Z_\delta(X; q) = \mathcal{Z}_\delta(X; q, q)$*. On every left Specht multiplicity space, identified with* S^λ*,*

$$\ker((F_k \delta F_k)|_{F_k S^\lambda}) = F_k \ker \delta,$$
$$\operatorname{rank}((F_k \delta F_k)|_{F_k S^\lambda}) = \sum_{j=0}^{k} [s_\lambda q^j](H_a[H_b]H_a - Z_\delta(X; q)).$$

For $a \geq 3$*, the inverse on the image has norm at most* $(\gamma_a \min_t w_a(t))^{-1}$*.*

Proof. On the old rows, decompose the right outer factor $\mathbb{C}[S_a]$ of the wreath model of Subsection S5.2 by its type α, and grade only by I_{old}. The left multiplicity of V_α is then the wreath induction of the tensor power of the inner coinvariant module, tensored with V_α, so its graded characteristic is $s_\alpha[H_b(X; u)]$. The new-row regular module contributes left type s_β paired with its right V_β. Thus a fixed pair of right types supplies the left multiplicity factor $s_\alpha[H_b(X; u)]s_\beta$ after induction to S_N.

By Theorem S6.1 applied to (S6.12), the right kernel is the diagonal V_γ specified by $\mathcal{E}_a$. On it, $D_{\mathrm{pair}} = L_a^{UV}$, whose graded trace is $\mathfrak{f}_\gamma(q)$ by (S5.4). With the old inner degree retained independently, these traces prove (S6.15). The full intermediate degree has characteristic $H_a[H_b]H_a$. Subtracting the kernel gives the rank. Finally the kernel reduces C, and the lower bound of Proposition S6.3 applies to the two commuting S_a actions, so the quadratic-form proof of Theorem S5.1 gives the compression and inverse assertions. □

For an expansion with no Kronecker notation, put $\theta_p = (a-p, 1^p)$ for $0 \le p \le a-1$. Equation (S6.15) becomes

$$\begin{aligned}\mathcal{Z}_\delta(X;u,q) = &\sum_{\substack{p,r\ge 0\\ p+r\le a-1}} \mathfrak{f}_{\theta_{p+r}}(q)\, s_{\theta_p}[H_b(X;u)]\, s_{\theta_r} \\ &+ \sum\nolimits_{\alpha \text{ nonhook}} \mathfrak{f}_\alpha(q)\big(s_\alpha[H_b(X;u)]h_a + h_a[H_b(X;u)]s_\alpha\big) \\ &+ q^{c_a} \sum\nolimits_{\alpha \text{ nonhook}} s_\alpha[H_b(X;u)]s_{\alpha'}.\end{aligned}$$

Because the two input actions are distinguished, the sum over all nonhooks counts both orientations of a distinct conjugate pair.

S6.9.1. *Intermediate and final cutoffs.* Theorem S6.13 controls the loss under cutoffs of the intermediate degree C of (S6.14), whereas the comparison on $b+1$ rows uses cutoffs of the final column degree $B_{a,b+1} = \sum_c L^{(c)}_{b+1} + L_a^{\text{whole columns}}$ of Section S5.3. The two cutoffs already disagree on the global sign representation: for every $a \ge 2$ and odd $b \ge a$,

$$\delta = a(a-1) > 0, \qquad C = a\binom{b}{2} + 2\binom{a}{2}, \qquad B_{a,b+1} = a\binom{b+1}{2}, \tag{S6.16}$$

and $B_{a,b+1} - C = a(b-a+1) > 0$. Indeed, the old and new column swaps both act by sign, while whole-column swaps on all $b+1$ rows act trivially. At $k = C$ the sign line lies in the intermediate cutoff, where the compressed loss has rank one, but not in the final B-cutoff, where the compressed gain has rank zero; no map from the first cutoff space into the second is injective. The comparison of P and Q itself is consistent with this family, since the final row sign degree is $A_{a,b+1} = (b+1)\binom{a}{2} + \mathbf{1}_{\{a \text{ odd}\}}\binom{b+1}{2}$ and

$$B_{a,b+1} - A_{a,b+1} = \begin{cases} a(b+1)(b-a+1)/2, & a \text{ even}, \\ (a-1)(b+1)(b-a)/2, & a \text{ odd}, \end{cases}$$

which is nonnegative. A proof must therefore keep the accumulated terms of the exchange recurrence rather than match isolated increments at a common cutoff.

S6.9.2. *The sharp lower bound for pair transfers.* Since the loss enters $B_{a,b+1}$ with a negative sign, a comparison needs maps out of its support that are bounded below. For a column transposition the loss is supported where both the old whole-column swap and the new-row swap act negatively, and selecting the first negative old row gives a transfer with a uniform lower bound.

Let $\mathcal{V}$ be a unitary representation of S_{2b+2}, with old rows (x_i, y_i) for $0 \le i < b$ and new row (z, w). Put

$$g_i = (x_i\, y_i), \quad g_b = (z\, w), \quad e_i^{\pm} = \frac{I \pm g_i}{2}, \quad O = \frac{I - \prod_{i<b} g_i}{2}, \quad P = Oe_b^-.$$

For $D \in \{O, P\}$, define

$$F_i^D = De_i^- \prod\nolimits_{j<i} e_j^+, \qquad \tau_i = (y_i\, z), \qquad T_D = \sum\nolimits_{i<b} \tau_i F_i^D. \tag{S6.17}$$

Here O projects onto odd old parity, and F_i^D projects onto the vectors of $D\mathcal{V}$ whose first negative old row is i; these projections are mutually orthogonal with sum D. Thus $T_D = T_D D$. For $b = 0$, the empty product makes $O = P = 0$. Let Z_i be the orthogonal antisymmetrizer on the three cells $\{x_i, y_i, z\}$. After the projection O, the terms of T_D for distinct i have orthogonal outputs, so the bound reduces to a computation in the copy of S_3 generated by g_i and τ_i, whose sign projector is Z_i.

Theorem S6.14 (Lower bound for pair transfers)**.** *For every $b \geq 0$, every unitary representation $\mathcal{V}$ of S_{2b+2}, and $D \in \{O, P\}$,*

$$T_D^* T_D - \tfrac{1}{4} D = T_D^*(I - O)T_D + \tfrac{3}{4} \sum_{i<b} F_i^D Z_i F_i^D \succeq 0. \tag{S6.18}$$

In particular, $\|T_D v\| \geq \frac{1}{2}\|v\|$ for $v \in D\mathcal{V}$. The constant $1/2$ is optimal for each $b \geq 2$, including on the loss space $P\mathcal{V}$. For $b = 1$ the transfer is an isometry on its source; for $b = 0$ the source is zero.

Proof. Write $v_i = F_i^D v$. Since v_i has odd old parity and is negative under g_i, the product of the other old row swaps fixes it. The transposition τ_i commutes with those swaps, so

$$O\tau_i v_i = e_i^- \tau_i v_i = (e_i^- \tau_i e_i^-) v_i. \tag{S6.19}$$

This output is positive under every old row swap preceding i and negative under g_i. Outputs for distinct i are therefore orthogonal.

For the local calculation, set $s = g_i$, $t = \tau_i$, and $e = (I - s)/2$. The transpositions s, t generate S_3, whose sign projector is $Z = \frac{1}{6}(I - s - t - sts + st + ts)$. Direct expansion gives

$$ete = \tfrac{1}{2}e - \tfrac{3}{2}Z, \qquad (ete)^2 = \tfrac{1}{4}e + \tfrac{3}{4}Z.$$

Using (S6.19) and the orthogonality of these outputs,

$$\|T_D v\|^2 = \|(I - O)T_D v\|^2 + \tfrac{1}{4}\|Dv\|^2 + \tfrac{3}{4}\sum_{i<b} \|Z_i F_i^D v\|^2,$$

which is (S6.18). The witness after this proof attains $1/2$ for $b = 2$. For $b > 2$, place its six active cells in the first two old rows and the new row. In $\mathrm{Ind}_{S_6 \times S_{2b-4}}^{S_{2b+2}}(V_{(2,2,2)} \boxtimes \mathbf{1})$, the identity-coset subspace contains an isometric copy of the witness. Every spectator row swap acts positively. All terms with $i \geq 2$ vanish, and the two surviving transfers preserve this copy. The same ratio is therefore attained for every $b \geq 2$. When $b = 1$, there is a single term and T_D is a transposition on its source. □

For $b = 2$ the following vector attains equality. In a two-column grid with two old rows and one new row, label the cells $0, \ldots, 5$ in row order and put $g_0 = (01)$, $g_1 = (23)$, $g_2 = (45)$. Define

$$F_0 = \tfrac{1}{8}(I - g_0)(I + g_1)(I - g_2), \qquad F_1 = \tfrac{1}{8}(I + g_0)(I - g_1)(I - g_2), \qquad P = F_0 + F_1,$$

and $T = (14)F_0 + (34)F_1$, which is T_D with $D = P$. In the sign-twisted permutation representation on three-subsets of six cells, take, in lexicographic subset order,

$$v = (-1, 1, 0, 0, 1, 0, 0, -1, -1, 1, -1, 1, 1, 0, 0, -1, 0, 0, -1, 1).$$

Direct integer multiplication gives

$$Pv = v, \qquad T^*Tv = \tfrac{1}{4}v, \qquad \|v\|^2 = 12, \qquad \|Tv\|^2 = 3. \tag{S6.20}$$

The transposition class sum acts on v by -3. Vertical Pieri decomposes this module into the four conjugates of $(6), (5,1), (4,2), (3,3)$, with distinct class-sum eigenvalues $-15, -9, -5, -3$. Thus v lies in the irreducible type $(2,2,2)$. By (S6.20), $\|Tv\| = \frac{1}{2}\|v\|$, the lower bound of Theorem S6.14.

The theorem applies to the loss, because each loss projector is an instance of P. For a column transposition $t = (c\,d)$, take for x_i, y_i the cells of columns c, d in old row i, and for z, w those in the new row. Then $\prod_{i<b} g_i = U_t$ and $g_b = V_t$, so the loss projector $P_t = \frac{1}{4}(I - U_t)(I - V_t)$ is the projection P, and Equation (S6.12) reads $\delta = 4\sum_t w_a(t) P_t$. Let T_t denote T_D of (S6.17) with $D = P_t$. Theorem S6.14 gives

$$\sum_t w_a(t)\|T_t P_t v\|^2 \geq \tfrac{1}{4}\sum_t w_a(t)\|P_t v\|^2 = \tfrac{1}{16}\langle v, \delta v\rangle.$$

Hence the map $v \mapsto (w_a(t)^{1/2} T_t v)_t$ into the direct sum has kernel exactly $\ker \delta$.

S7. Positive averages over final cutoffs

The conjecture asks for $[s_\lambda q^k]F_{a,b} \geq 0$ at every cutoff k, and the sum over all cutoffs is $[s_\lambda]F_{a,b}(X;1)$. Bergeron gave a closed formula for $F_{a,b}(X;1)$ and proved that it is a polynomial in h_1, h_2, e_2 with nonnegative integer coefficients, hence Schur positive (Bergeron, 2017, Proposition 7 and Appendix B). In this basis the average over cutoffs is a Gram operator, and residue classes of cutoffs carry equal parts of the total.

Fix $2 \leq a \leq b$ and $N = ab$, and put $D_P = b\binom{a}{2} + \binom{b}{2}$ and $D_Q = a\binom{b}{2} + \binom{a}{2}$. Write $F_{a,b}(X;q) = \sum_{k=0}^{D_Q-1} F_k(X)q^k$. Since $D_P \leq D_Q$, the divided difference has degree at most $D_Q - 1$. For $E = (\mathbb{C}^d)^{\otimes N}$ with $d \geq N$, let Π_λ be the full Schur–Weyl isotypic projection, and define

$$Z(f) = \sum\nolimits_{\lambda \vdash N} \frac{[s_\lambda]f}{d_\lambda}\Pi_\lambda, \qquad Z_k = Z(F_k).$$

Thus $Z_k \succeq 0$ is equivalent to $[s_\lambda]F_k \geq 0$ for every λ at that cutoff.

Proposition S7.1 (Cutoff average)**.** *For $2 \leq a \leq b$,*

$$F_{a,b}(X;1) = h_1^{N-2b} \sum\nolimits_{j=0}^{b} C_j(a,b) h_2^{b-j} e_2^j, \tag{S7.1}$$

where the nonnegative integers $C_j(a,b)$ satisfy

$$\begin{aligned} 2C_j(a,b) = {} & ab(b-a)\binom{b-1}{j-1} + a(a-1)\mathbf{1}_{\{j \text{ odd}\}}\binom{b}{j} \\ & - b(b-1)\sum\nolimits_{\ell \text{ odd}} \binom{a}{\ell}\binom{b-a}{j-\ell}. \end{aligned} \tag{S7.2}$$

Out-of-range binomial coefficients are zero. Choose disjoint transpositions $\tau_i = (2i-1, 2i)$ for $1 \leq i \leq b$ and let ρ denote the permutation action on E. With the orthogonal projections

$$Q_j = \prod\nolimits_{i=1}^{b-j} \frac{I + \rho(\tau_i)}{2} \prod\nolimits_{i=b-j+1}^{b} \frac{I - \rho(\tau_i)}{2}$$

the Gram identity reads

$$\frac{1}{D_Q}\sum\nolimits_{k=0}^{D_Q-1} Z_k = \frac{1}{D_Q N!}\sum\nolimits_{g \in S_N}\sum\nolimits_{j=0}^{b} C_j(a,b)\rho(g)Q_j\rho(g)^* \succeq 0. \tag{S7.3}$$

Proof. Only the identity and transposition cycle types contribute to the first derivative at $q = 1$ in the power-sum formula for H_n, so

$$H_n'(1) = \tfrac{c_n}{2}(p_1^n - p_2 p_1^{n-2}).$$

Plethystic differentiation, with $\psi_r(q) = q^r$, therefore gives

$$(H_b[H_a])'(1) = \tfrac{bc_a}{2}(p_1^N - p_2 p_1^{N-2}) + \tfrac{c_b}{2}(p_1^N - p_2^a p_1^{N-2a}).$$

Subtracting in the order $F_{a,b}(1) = Q'_{a,b}(1) - P'_{a,b}(1)$ yields

$$\begin{aligned} 4F_{a,b}(1) = {} & (b-a)(a-1)(b-1)p_1^N - ab(b-a)p_2 p_1^{N-2} \\ & + b(b-1)p_2^a p_1^{N-2a} - a(a-1)p_2^b p_1^{N-2b}. \end{aligned}$$

Substituting $p_1^2 = h_2 + e_2$ and $p_2 = h_2 - e_2$ proves (S7.1)–(S7.2), which is Bergeron's formula (Bergeron, 2017, (23)) in this basis. All three prefactors in (S7.2) are even, so every C_j is integral.

The last term of $2C_j(a,b)$ carries a minus sign, so we prove $C_j \geq 0$ by writing C_j as a multiple of the expectation of a nonnegative quantity. Mark a elements of a b-element set and sample an ordered j-tuple without replacement. Pair consecutive entries, leaving a singleton when j is odd.

Let Y count pairs with exactly one marked endpoint, together with a marked singleton if present; let U count all sampled marked elements. Then

$$Y - \mathbf{1}_{\{U \text{ odd}\}} = 2\lfloor Y/2 \rfloor, \qquad \mathbb{E}Y = \frac{a(b-a)j + a(a-1)\mathbf{1}_{\{j \text{ odd}\}}}{b(b-1)}.$$

The hypergeometric formula for $\mathbb{P}(U \text{ odd})$ turns (S7.2) into

$$C_j(a,b) = b(b-1)\binom{b}{j}\mathbb{E}\lfloor Y/2 \rfloor \geq 0.$$

The weighted trace of Q_j is $\mathrm{Tr}(Q_j X^{\otimes N}) = h_1(X)^{N-2b} h_2(X)^{b-j} e_2(X)^j$. Averaging its conjugates over S_N replaces each Specht block by its trace divided by d_λ. Applying this averaging to (S7.1) proves (S7.3). Each summand is a Gram operator because $Q_j = Q_j^* Q_j$. □

The first coefficients are $C_0 = C_1 = C_2 = 0$ and $C_3 = a(a-1)b(b-1)(b-a)/3$. Thus for $b > a$,

$$F_{a,b}(1) - \frac{a(a-1)b(b-1)(b-a)}{3} h_1^{N-2b} h_2^{b-3} e_2^3 \in \mathbb{N}\{s_\lambda\}.$$

The average also has a refinement that retains the cutoff modulo r.

Proposition S7.2 (Residue averages)**.** *Let $2 \leq a \leq b$, and let $r \geq 2$ satisfy $a \bmod r \in \{0,1\}$ and $b \bmod r \in \{0,1\}$. Then $[r]_q$ divides every Schur coefficient of $F_{a,b}(X;q)$ in $\mathbb{Z}[q]$, and for every $0 \leq c < r$,*

$$\sum_{\substack{0 \leq k < D_Q \\ k \equiv c \pmod r}} F_k = \frac{1}{r} F_{a,b}(1), \qquad \sum_{\substack{0 \leq k < D_Q \\ k \equiv c \pmod r}} Z_k = \frac{1}{r} Z(F_{a,b}(1)) \succeq 0. \tag{S7.4}$$

Every pair a, b satisfies the hypothesis for $r = 2$, so the sums over even and odd cutoffs are always equal and Schur positive.

Proof. Let $\zeta \neq 1$ be an rth root of unity, of order t. The hypotheses also hold modulo t. For $n = tu + \epsilon$ with $\epsilon \in \{0,1\}$, we let $q \to \zeta$ in the power-sum formula

$$H_n(X;q) = (q;q)_n \sum\nolimits_{\nu \vdash n} \frac{p_\nu(X)}{z_\nu \prod_i (1 - q^{\nu_i})}$$

and obtain $H_n(X;\zeta) = p_t^u p_1^\epsilon$. Indeed, the numerator has a zero of order u, and only $\nu = (t^u, 1^\epsilon)$ has u denominator factors vanishing at ζ; their ratio, including z_ν, has limit one. Write $a = tu + \epsilon$ and $b = tv + \epsilon'$. The Adams convention gives $\psi_t(H_a)|_{q=\zeta} = H_a(X^t;1) = p_t^a$, so both nested plethysms specialize to the same power-sum product

$$p_t^{av + u\epsilon'} p_1^{\epsilon\epsilon'} = p_t^{\lfloor N/t \rfloor} p_1^{N \bmod t}.$$

Hence $F_{a,b}(\zeta) = 0$. Its integral Schur coefficients are divisible by the monic polynomial $[r]_q$, whose roots are all nontrivial rth roots. Reducing this divisibility modulo $q^r - 1$ shows that all residue sums are equal, and their total is $F_{a,b}(1)$, which proves the first identity in (S7.4); the second follows by linearity of Z and Proposition S7.1. □

Part 4. Polarization and the expansion hypothesis

S8. Quantitative power polarization

The norm bound in Paper III, Lemma 2.1, quantifies the polarization steps in the proof of Paper III, Theorem 1.3. Its boundary case determines the threshold in that argument. We record the quantitative polarization that the bound gives and show for $a = 6$ that symmetric inputs do not lower the threshold of the last step. Write $x^{\underline{L}} = x(x-1)\cdots(x-L+1)$, with empty product one.

The injectivity in that lemma is the $\mathfrak{sl}_2$ argument behind the full-tensor polarization of the ordinary sixth-power argument (OpenAI, 2026, Lemma 2.5): with $W_j = (\mathrm{Sym}^j V)^{\otimes a}$ and $z(v) = v^{\otimes a}$,

$$W_j = \sum\nolimits_{v\in V} z(v)^L W_{j-L}, \qquad j \ge aL. \tag{S8.1}$$

Multiplication by $z(v)^L$ commutes with S_a, so this decomposition holds in every outer S_a-isotypic component. Taking $L = 1$ shows, as in OpenAI (2026, Corollary 2.6), that the graded module $\bigoplus_{j\ge 0} \mathrm{Hom}_{S_a}(S^\nu, W_j)$ over the algebra generated by the diagonal tensors $z(v)$ is generated in degrees less than a.

The same norm bound makes this polarization quantitative for the Fischer inner product, which is the L^2 inner product for independent standard complex Gaussian coordinates. For $f(X_1,\dots,X_a)$ homogeneous of degree j in each X_i, set $D_i = Y\cdot\partial_{X_i}$. Then

$$\Big\|\prod\nolimits_{i=1}^{a} D_i^L f\Big\|^2 \ge (L!)^a\, j^{\underline{aL}}\, \|f\|^2, \qquad j \ge aL. \tag{S8.2}$$

The power lemma applies to each group of derivatives. Before the ith group, the degree gap is $j-(i-1)L \ge L$; multiplying these bounds proves (S8.2). To recover (S8.1), pair its right side with a polynomial $f \in W_j$ using the Fischer inner product. Orthogonality for every v is equivalent, by the multiplication–differentiation adjunction, to $\prod_i D_i^L f = 0$ as a polynomial in Y. The estimate forces $f = 0$, so the orthogonal complement vanishes. The same proof allows extra coefficient spaces.

Set $r = a-1$. The last step of the surjectivity proof applies the r blocks $(D_{x,v_i})^r$ in turn. Their norm bounds combine into one estimate for the composition.

Corollary S8.1 (Quantitative common-power estimate)**.** *On polynomials of degree b in each v_i, initially independent of x, if $b \ge r^2$, then*

$$\Big\|\prod\nolimits_{i=1}^{r} (D_{x,v_i})^r f\Big\|^2 \ge (r!)^r\, b^{\underline{r^2}}\, \|f\|^2.$$

Proof. Multiply the r bounds (2.1). The falling factorials have consecutive, disjoint ranges and combine to $b^{\underline{r^2}}$. □

For $a = 6$, the last step of that proof applies five blocks $(D_{x,v_i})^5$ (Corollary S8.1), and the threshold 25 cannot be lowered to 24 by asserting their injectivity on inputs symmetric in $v_1,\dots,v_5$. Indeed, on $V = \mathbb{C}^5$, the polynomial $f(v_1,\dots,v_5) = \det(v_1,\dots,v_5)^{24}$ is nonzero and symmetric in the five vector variables, but $\prod_{i=1}^5 (D_{x,v_i})^5 f = 0$. In each Leibniz term, 25 copies of the same vector x are inserted into 24 determinant factors; one factor receives two copies and vanishes.

S9. The current space and the expansion hypothesis

Paper III, Theorem 4.1, reduces full q-Foulkes positivity to an expansion hypothesis for a specified space of current maps. We restate the hypothesis in the notation of the tests below. These tests use homogeneous generators and the standard certificates for a matrix space with no shrunk subspace.

Fix $b \ge a \ge 2$, put $N = ab$ and $G = S_N$, and let $\mathcal{P} = M_{a,b}$ and $\mathcal{Q} = M_{b,a}$ be the nested coinvariant modules of Paper III, Equation (3.1). Put

$$U = \mathcal{P}_0 = \mathrm{Ind}_{S_a\wr S_b}^{S_N} \mathbf{1}, \qquad V = \mathcal{Q}_0 = \mathrm{Ind}_{S_b\wr S_a}^{S_N} \mathbf{1}, \qquad D_P = bc_a + c_b, \quad D_Q = ac_b + c_a.$$

Use the centered Fischer realizations on $W_0 = \{w \in \mathbb{C}^N : \sum_i w_i = 0\}$ from Paper III, Section B. For an equivariant homogeneous seed $\sigma_r : U \to \mathcal{Q}_r$, let $\mathcal{A}_j(\sigma_r)$ be the coefficient map of Paper III, Equation (4.1). It has degree shift $r \geq 0$. The current space is

$$\mathscr{M}_{a,b} = \operatorname{span}_{\mathbb{Q}}\{\mathcal{A}_j(\sigma_r) : \sigma_r \in \operatorname{Hom}_{\mathbb{Q}[S_N]}(U, \mathcal{Q}_r),\ 0 \leq r \leq D_Q,\ 0 \leq j \leq D_P\}. \tag{S9.1}$$

The free factors and harmonic projections that define these maps act before passage to an isotypic component. All modules and maps have rational forms; subspaces below are taken over $\mathbb{C}$.

Write $\mathcal{E}^P_\lambda$ and $\mathcal{E}^Q_\lambda$ for the graded multiplicity spaces in $\mathcal{P}$ and $\mathcal{Q}$, each of total dimension d_λ, and let $\mathscr{M}_\lambda$ be the induced complex matrix space. For $W \subseteq \mathcal{E}^P_\lambda$, put $\mathscr{M}_\lambda(W) = \operatorname{span}\{Mv : M \in \mathscr{M}_\lambda,\ v \in W\}$. A label is neutral if the two multiplicity spaces have equal dimensions in every degree. The expansion hypothesis is

$$\dim \mathscr{M}_\lambda(W) \geq \dim W \quad \text{for every nonneutral } \lambda \text{ and every graded } W \subseteq \mathcal{E}^P_\lambda. \tag{S9.2}$$

Here graded means $W = \bigoplus_k (W \cap \mathcal{E}^P_{\lambda,k})$. A subspace violating this inequality is called *shrunk*. Paper III proves that (S9.2) implies $F_{a,b} \succeq_s 0$; we do not claim the converse for this current family.

S10. Certificates for the expansion hypothesis

The expansion hypothesis concerns graded subspaces only, whereas the usual expansion criterion for matrix spaces and its capacity formulation concern all subspaces. Homogeneity reconciles the two, because a shrunk subspace can be replaced by a graded witness.

Lemma S10.1 (Graded witnesses suffice)**.** *Let $\mathscr{M} \subseteq \operatorname{Hom}(E, F)$ be spanned by homogeneous maps between finite-dimensional graded spaces. If $\dim \mathscr{M}(W) < \dim W$ for some subspace $W \subseteq E$, then the same strict inequality holds for a graded subspace of E.*

Proof. Put $Z = \mathscr{M}(W)$ and use the increasing degree filtrations on W and Z. Identify their associated graded spaces with the spaces of leading homogeneous components in E and F. For a homogeneous generator M, the image of a leading component is either zero or the leading component of $Mw \in Z$. Hence

$$\mathscr{M}(\operatorname{gr} W) \subseteq \operatorname{gr} Z, \qquad \dim \mathscr{M}(\operatorname{gr} W) \leq \dim Z < \dim W = \dim \operatorname{gr} W.$$

Thus $\operatorname{gr} W$ is the required graded witness. □

S10.1. **Capacity and determinant certificates.** On each block, the expansion hypothesis is equivalent to positive capacity of a completely positive map, and nonzero determinant coefficients certify positive capacity. Choose a finite homogeneous spanning family $M_1, \dots, M_\ell$ of $\mathscr{M}_\lambda$ in orthonormal multiplicity bases, put $n = d_\lambda$, and define

$$\Phi_\lambda(X) = \sum\nolimits_{i=1}^{\ell} M_i X M_i^*, \qquad \operatorname{cap}(\Phi_\lambda) = \inf_{X \succ 0,\ \det X = 1} \det \Phi_\lambda(X). \tag{S10.1}$$

Lemma S10.1 shows that a shrunk subspace has a graded witness. The operator-scaling characterization therefore gives

$$(\text{S9.2}) \quad \Longleftrightarrow \quad \operatorname{cap}(\Phi_\lambda) > 0 \quad \text{for every nonneutral } \lambda$$

(Garg et al., 2020, Theorem 1.4 and Section 2.1). Indeed, for $X \succeq 0$ with range W, $\operatorname{im} \Phi_\lambda(X) = \mathscr{M}_\lambda(W)$. Complete positivity of Φ_λ follows from its displayed form; positive capacity is the additional requirement. The proof of Paper III, Theorem 4.1, uses only degree tails, not operator scaling.

The quantum permanent gives a sum-of-squares certificate for an invertible member of $\mathscr{M}_\lambda$. Write $\operatorname{QP}(\Phi_\lambda)$ for Gurvits's quantum permanent of the Choi matrix of Φ_λ. With independent standard circular complex Gaussians z_i, normalized by $\mathbb{E}|z_i|^2 = 1$, it is

$$\operatorname{QP}(\Phi_\lambda) = \mathbb{E}\left|\det\Big(\sum\nolimits_i z_i M_i\Big)\right|^2 \geq 0 \tag{S10.2}$$

(Gurvits, 2004, Theorem 3.8). Strict positivity is equivalent to the existence of an invertible member of the Kraus span, and hence implies (S9.2) on that block. A normalized current family with a nonzero determinant polynomial supplies such a member by specialization. The quantum permanent also gives the quantitative bound $\operatorname{cap}(\Phi_\lambda) \geq \operatorname{QP}(\Phi_\lambda)/n!$.

A positive quantum permanent is sufficient for expansion but not necessary. The skew-symmetric example in Section S10.4 has positive capacity and zero quantum permanent. Matrix-valued coefficients recover the full expansion criterion. For $d \geq 1$, let $\mathcal{D}_d(Y) = \operatorname{tr}(Y) I_d$ and put $\operatorname{QP}_d(\Phi_\lambda) = \operatorname{QP}(\Phi_\lambda \otimes \mathcal{D}_d)$. The Kraus matrices of this map are $M_i \otimes E_{uv}$ for all $1 \leq u, v \leq d$, with the auxiliary space placed in degree zero. Proposition S10.2 gives

$$\text{(S10.3)} \qquad \text{(S9.2)} \iff \text{for every nonneutral } \lambda, \text{ some } d \geq 1 \text{ satisfies } \operatorname{QP}_d(\Phi_\lambda) > 0.$$

Thus the required certificate is a nonzero determinant coefficient at some matrix size. Nonnegativity alone is automatic and proves no expansion.

S10.2. **Quantum permanents and matrix-valued coefficients.** The certificates above rest on an expression of the quantum permanent through determinant coefficients, which we now prove. For a completely positive map $\Phi(X) = \sum_i M_i X M_i^*$ on $n \times n$ matrices, its Choi matrix is $C_\Phi = \sum_{u,v} E_{uv} \otimes \Phi(E_{uv})$. We use the unnormalized invariant $\operatorname{QP}(\Phi) = \operatorname{QP}(C_\Phi)$, so that $\operatorname{QP}(\mathrm{id}_{M_n}) = n!$. The determinant coefficients make its strict positivity test explicit.

Proposition S10.2 (Quantum-permanent certificates)**.** *Let $M_1, \dots, M_\ell$ be complex $n \times n$ matrices, where $n \geq 1$, and put $\Phi(X) = \sum_i M_i X M_i^*$. If*

$$\det\Big(\sum\nolimits_i z_i M_i\Big) = \sum\nolimits_{|\alpha|=n} c_\alpha z^\alpha, \qquad \alpha! = \prod\nolimits_i \alpha_i!,$$

then

$$\text{(S10.4)} \qquad \operatorname{QP}(\Phi) = \sum\nolimits_{|\alpha|=n} \alpha! |c_\alpha|^2, \qquad \operatorname{cap}(\Phi) \geq \frac{\operatorname{QP}(\Phi)}{n!}.$$

In particular, $\operatorname{QP}(\Phi) > 0$ if and only if $\operatorname{span}(M_1, \dots, M_\ell)$ contains an invertible matrix. For $\mathcal{D}_d(Y) = \operatorname{tr}(Y) I_d$ and $\operatorname{QP}_d(\Phi) = \operatorname{QP}(\Phi \otimes \mathcal{D}_d)$,

$$\text{(S10.5)} \qquad \operatorname{cap}(\Phi) > 0 \iff \operatorname{QP}_d(\Phi) > 0 \text{ for some } d \geq 1, \qquad \operatorname{cap}(\Phi) \geq \left(\frac{\operatorname{QP}_d(\Phi)}{(nd)!\, d^{nd}} \right)^{1/d}.$$

The constant $n!$ in (S10.4) is optimal.

Proof. Gurvits's Gaussian formula (Gurvits, 2004, Theorem 3.8) identifies the quantum permanent with the expectation in (S10.2). Gaussian monomials satisfy $\mathbb{E}(z^\alpha \overline{z^\beta}) = \mathbf{1}_{\{\alpha=\beta\}} \alpha!$, proving the first formula in (S10.4). Its strict positivity is equivalent to the determinant polynomial being nonzero, hence to an invertible scalar combination.

For $X \succ 0$, replace every Kraus matrix by $M_i X^{1/2}$ and denote the resulting map by Φ_X. Determinant multiplicativity gives $\operatorname{QP}(\Phi_X) = \det(X) \operatorname{QP}(\Phi)$. Gurvits's bound (Gurvits, 2004, Corollary 3.5) gives

$$\det(X) \operatorname{QP}(\Phi) = \operatorname{QP}(\Phi_X) \leq n! \det \Phi_X(I_n) = n! \det \Phi(X).$$

Taking the infimum over $\det X = 1$ proves the capacity bound. The identity map has capacity one and quantum permanent $n!$, so the constant is optimal.

The map $\Phi \otimes \mathcal{D}_d$ has Kraus span $\operatorname{span}(M_1, \dots, M_\ell) \otimes M_d(\mathbb{C})$. By the no-shrunk-subspace characterization and the matrix-amplification theorem (Garg et al., 2020, Theorem 1.4 and Corollary 2.16), positive capacity holds exactly when this span contains an invertible member for some d. The first part of the proposition therefore proves the equivalence in (S10.5). For its bound, apply the preceding determinant inequality in dimension nd to $X \otimes I_d$, where $\det X = 1$. Since

$$(\Phi \otimes \mathcal{D}_d)(X \otimes I_d) = \Phi(X) \otimes d I_d,$$

we obtain $\mathrm{QP}_d(\Phi) \le (nd)!\, d^{nd} \det \Phi(X)^d$. Take dth roots and then the infimum. □

Equivalently, $\mathrm{QP}_d(\Phi)$ is $\mathbb{E}|\det(\sum_i M_i \otimes G_i)|^2$, where the entries of the $d \times d$ matrices G_i are independent standard circular complex Gaussians. Its sum-of-squares expansion uses all coefficients of this determinant polynomial. Tensoring with the identity map would retain only the span of $M_i \otimes I_d$ and would not supply this criterion. For the homogeneous current maps, the auxiliary factor has trivial group action and degree zero, so every amplified map still preserves the direction of the degree filtration.

S10.3. **A Gram certificate.** Unlike the quantum-permanent certificates, the next certificate uses only the two Gram sums. It requires exact normalization on one side and a strict error bound on the other. Its proof sharpens the tolerance needed here by using the zero trace of the normalization error; compare Garg et al. (2020, Theorem 2.12).

Proposition S10.3 (A nearly balanced Gram certificate)**.** *Let $M_1, \dots, M_\ell$ span a space of maps between two n-dimensional Hermitian spaces. Suppose that invertible matrices L, R give matrices $B_i = LM_iR$ with*

$$\sum\nolimits_i B_i B_i^* = I_n. \tag{S10.6}$$

For $n \ge 2$, assume also

$$\Big\|\sum\nolimits_i B_i^* B_i - I_n\Big\|_{\mathrm{F}} < \frac{2}{\sqrt{n}}. \tag{S10.7}$$

Then the matrix space has no shrunk subspace. For $n = 1$, (S10.6) alone suffices.

Proof. Invertible left and right changes preserve the existence of shrunk subspaces. Work with the B_i, and set $E = \sum_i B_i^* B_i - I_n$. Row normalization gives $\operatorname{tr} E = 0$. Suppose an r-dimensional subspace W has its images contained in an s-dimensional subspace Z, with $s < r$. Row normalization makes the total image the whole codomain, so $1 \le r \le n-1$. For the orthogonal projection P_W onto W,

$$0 \preceq \sum\nolimits_i B_i P_W B_i^* \preceq I_n, \qquad \operatorname{im}\Big(\sum\nolimits_i B_i P_W B_i^*\Big) \subseteq Z.$$

Consequently

$$\operatorname{tr}(P_W E) = \operatorname{tr}\Big(\sum\nolimits_i B_i P_W B_i^*\Big) - r \le s - r \le -1.$$

Since $\operatorname{tr} E = 0$, the Frobenius Cauchy–Schwarz inequality gives

$$1 \le |\operatorname{tr}((P_W - (r/n) I_n) E)| \le \sqrt{\frac{r(n-r)}{n}}\, \|E\|_{\mathrm{F}}.$$

Thus $\|E\|_{\mathrm{F}}^2 \ge n/(r(n-r)) \ge 4/n$, contradicting (S10.7). For $n = 1$, row normalization makes at least one map nonzero. □

The proof also permits the stronger squared-norm threshold $n/\lfloor n^2/4 \rfloor$ when n is odd. The changes L, R need not preserve degree, because absence of shrunk subspaces is invariant under invertible coordinate changes. These tests certify the hypothesis of Paper III, Theorem 4.1; they are not asserted to hold for the sixth-power current spaces.

S10.4. **Scalar determinants and Gram positivity.** Two 3×3 examples show that neither the matrix amplification nor the normalization of the Gram sums can be dropped. Let E_{ij} be the 3×3 matrix units and put

$$M_1 = E_{12} - E_{21}, \qquad M_2 = E_{13} - E_{31}, \qquad M_3 = E_{23} - E_{32}.$$

Both Gram sums are $2I_3$, so dividing by $\sqrt{2}$ gives an exact certificate in Proposition S10.3. Nevertheless every scalar combination is an odd-dimensional skew-symmetric matrix and has determinant

zero. The example exhibits the familiar distinction between commutative and noncommutative rank (Garg et al., 2020, Example 1.3). For a direct matrix amplification, the choices

$$X_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad X_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

give $\det(\sum_i M_i \otimes X_i) = 1$. The CP map with Kraus matrices M_1, M_2, M_3 therefore has $\mathrm{QP}_1 = 0$ and $\mathrm{QP}_2 > 0$.

Positive definite Gram sums without normalization do not suffice. For

$$C_1 = E_{11}, \qquad C_2 = E_{12} + E_{21}, \qquad C_3 = E_{13} + E_{31},$$

both Gram sums are $\mathrm{diag}(3, 1, 1)$, but the entire matrix space sends $\mathrm{span}(e_2, e_3)$ into $\mathrm{span}(e_1)$, so this subspace is shrunk. These are finite matrix examples, not verifications of a Foulkes parameter.

S11. Exact block calculations

The expansion hypothesis is open in general, so we test it by exact calculation on individual multiplicity blocks. These calculations retain the final harmonic projection in the current construction. After the ground columns are removed, it remains open whether the residual maximal minors are nonvanishing on every nonneutral block.

S11.1. **The first two degrees and the quadratic rank condition.** In the range of the surjectivity theorem of Paper III, the canonical incidence map gives surjective current blocks in the first two degrees. In the next degree a seed of shift two adds freedom, and surjectivity becomes a rank condition on that seed. Use the free factors and harmonic block formula of Paper III, Proposition B.1 and Equation (B.5).

The ideals have no linear relations on W_0, so $\mathcal{P}_1 = \mathscr{P}_1(U)$ and $\mathcal{Q}_1 = \mathscr{P}_1(V)$. Let $\mu\colon U \to V$ be the canonical incidence map, whose nonzero entries correspond to transverse row and column partitions, and take $\sigma_0 = \mu$. Naturality gives

$$T_d^{0,0} = \mu, \qquad T_d^{1,1} = N^{-1} A_1^V(d)(I_{W_0^*} \otimes \mu).$$

For $b \geq (a-1)^2$ and $d \geq N$, the surjectivity theorem (Paper III, Theorem 1.3) makes both blocks surjective, because μ is a multiple of the multilinear component of (1.1) and $A_1^V(d) \succeq (d - N + 1)I$ by (B.3). In this range, if also $\sigma_1 = 0$, the truncation to degrees at most two has matrix

$$\begin{pmatrix} \mu & 0 & 0 \\ 0 & T_d^{1,1} & 0 \\ \sigma_2 & 0 & T_d^{2,2} \end{pmatrix}.$$

After solving its first two equations, the remaining degree-zero freedom is $\ker \mu$. Thus this truncation is surjective exactly when

$$\operatorname{im} T_d^{2,2} + \sigma_2(\ker \mu) = \mathcal{Q}_2. \tag{S11.1}$$

Equation (S11.1) is a remaining rank condition for the chosen seed.

S11.2. **Eliminating the ground columns.** All homogeneous seeds are available in (S9.1), so a shrunk graded subspace cannot meet domain degree zero. This observation also removes the columns of domain degree zero, the ground columns, from the quantum-permanent calculation.

Proposition S11.1 (Ground-column elimination)**.** *Fix a multiplicity block, and write* $E = \mathcal{E}_\lambda^P$, $F = \mathcal{E}_\lambda^Q$, $n = \dim E = \dim F$, $u = \dim E_0$, *and* $v = \dim F_0$. *Let* Π_k^E, Π_l^F *be the degree projections. Define*

$$\widehat{\mathscr{M}} = \mathrm{span}\{\Pi_l^F M \Pi_k^E : M \in \mathscr{M}_\lambda,\ k, l \geq 0\}, \qquad E_+ = \bigoplus_{k \geq 1} E_k, \quad F_+ = \bigoplus_{l \geq 1} F_l.$$

There is a matrix space $\mathscr{B} \subseteq \mathrm{Hom}(E_+, F_+)$ *such that*

$$\widehat{\mathscr{M}} = \{[\,C \mid B\,] : C \in \mathrm{Hom}(E_0, F),\ B \in \mathscr{B}\}, \tag{S11.2}$$

where B *has zero rows in* F_0*. Hypothesis* (S9.2) *on this block is equivalent to* $\dim \mathscr{B}(W) \geq \dim W$ *for every* $W \subseteq E_+$*.*

Choose a spanning family $B_1, \dots, B_s$ *of* $\mathscr{B}$ *in orthonormal bases. Let* $\widehat{\Phi}$ *have Kraus matrices consisting of these matrices, supported on* E_+*, and all matrix units from* E_0 *to* F*. Let the entries of all* $G_i \in M_d(\mathbb{C})$ *be independent standard circular complex Gaussians. For* $B_d = \sum_i B_i \otimes G_i$*, the quantum permanent satisfies*

$$\mathrm{QP}_d(\widehat{\Phi}) = (ud)!\,\mathbb{E}\det(B_d^* B_d) = (ud)! \sum_{\substack{I \subseteq \{1,\dots,(n-v)d\} \\ |I| = (n-u)d}} \mathbb{E}|\det(B_d)_I|^2. \tag{S11.3}$$

Consequently (S9.2) *holds on the original block if and only if this expression is positive for some* $d \geq 1$*. Empty determinants equal one; a sum with no admissible row set is zero.*

Proof. For a graded subspace $W \subseteq E$, homogeneity gives $\widehat{\mathscr{M}}(W) = \mathscr{M}_\lambda(W)$, because each degree block of a homogeneous generator sends a vector of W_k either to zero or to its image under that generator. Lemma S10.1 therefore shows that the two matrix spaces have the same expansion property. On E_0, the current coefficient with $j = 0$ is the seed itself, and those with $j > 0$ vanish. The seed spaces induce all of $\mathrm{Hom}(E_0, F_l)$, by semisimplicity. Separating the degree blocks makes these ground columns independent of the remaining columns, proving (S11.2). Nonnegative degree shifts force the latter columns to vanish in F_0.

If a subspace of E has a vector with nonzero E_0 component, the arbitrary matrices C send that vector onto all of F. A shrunk subspace for (S11.2) must therefore lie in E_+, where its image is exactly $\mathscr{B}(W)$. This proves the asserted expansion equivalence.

For $d = 1$, condition on $B = \sum_i z_i B_i$ and let C be an independent $n \times u$ standard circular Gaussian matrix. If B has deficient column rank, both $|\det[C \mid B]|^2$ and $\det(B^*B)$ vanish. Otherwise a unitary change of rows makes its last $n - u$ columns equal to $\binom{0}{R}$, with R square and $|\det R|^2 = \det(B^*B)$. The remaining determinant uses a $u \times u$ standard Gaussian matrix, whose squared determinant has expectation $u!$. Thus

$$\mathbb{E}_C|\det[C \mid B]|^2 = u!\det(B^*B).$$

This also holds for $u = 0$ and $u = n$ under the stated conventions. Apply the same calculation with nd rows and ud ground columns, and then Cauchy–Binet, to obtain (S11.3). Proposition S10.2 applied to $\widehat{\mathscr{M}}$ proves the final equivalence. □

The numerical quantum permanent in (S11.3) belongs to the displayed spanning family of $\widehat{\mathscr{M}}$; it is not identified with that of the original current family. The reduction preserves the existence of a positive amplified certificate. The residual matrices $B \in \mathscr{B}$ have $(n - v)$ rows and $(n - u)$ columns. For $a = 6$, ordinary Foulkes positivity gives $u \geq v$ for every $b \geq 6$, so there are enough rows. It does not establish the nonvanishing of any of these maximal minors.

S11.3. **A uniform certificate on the sign block.** The sign multiplicity is one. A single nonzero projected current therefore certifies expansion on this block, without amplification. Specht-ideal containment supplies such a current uniformly in b.

Proposition S11.2 (Sign-block expansion)**.** *Let* $a \geq 2$ *be even and* $b \geq a$*. On the sign block, the space* (S9.1) *contains a nonzero scalar current with* $j = 0$ *and seed degree*

$$r = \frac{ab(b-a)}{2} + \varepsilon_b \binom{a}{2}.$$

Hence its scalar quantum permanent is strictly positive. For $a = 6$, the domain and codomain sign degrees are $15b$ and $6\binom{b}{2} + 15\varepsilon_b$, respectively.

Proof. Use the harmonic fibers of Paper III, Section B, with $G = S_{ab}$, and write $\Delta_m(x) = \prod_{i<j}(x_j - x_i)$. For a domain partition π into b blocks of size a, set

$$p_\pi = \prod\nolimits_{R\in\pi} \Delta_a(w_R), \qquad A_\pi = \mathrm{Stab}_G(\pi).$$

This polynomial is harmonic in the domain fiber. It transforms under A_π by the restriction of the ambient sign character, because a is even, so block permutations have even ambient sign and leave the product unchanged. Thus the domain sign line is generated by

$$s = \sum\nolimits_{g\in G/A_\pi} \mathrm{sgn}(g)g(p_\pi \otimes e_\pi).$$

For a codomain partition κ into a blocks of size b, with block means $\eta_1, \dots, \eta_a$, its sign harmonic is

$$q_\kappa = \prod\nolimits_{K\in\kappa} \Delta_b(w_K)\Delta_a(\eta_1, \dots, \eta_a)^{\varepsilon_b}.$$

The inner and outer factors use orthogonal coordinate subspaces. Each is harmonic for its coinvariant ideal; their product transforms by the ambient sign. Consequently the full codomain sign vector t has components q_κ, with compatible signs. The degrees of p_π and q_κ give the stated value of r.

We claim that $p_\pi(D)t \neq 0$ for some π. The product of the inner Vandermondes in q_κ is a Specht polynomial of shape (a^b); the p_π are the Specht polynomials of shape (b^a). Dominance $(b^a) \trianglerighteq (a^b)$ gives the ideal containment

$$I^{\mathrm{sp}}_{(a^b)} \subseteq I^{\mathrm{sp}}_{(b^a)}$$

(Moustrou et al., 2021, Theorem 1). Restricting to W_0 and taking homogeneous components yields $q_\kappa = \sum_\pi p_\pi h_\pi$, with $\deg h_\pi = r$. These translation-invariant Vandermonde products remain nonzero on W_0. If all $p_\pi(D)q_\kappa$ vanished, Fischer adjunction would give $\langle q_\kappa, q_\kappa\rangle = 0$, a contradiction. This proves the claim.

Choose such a π and put $v = p_\pi(D)t$. Derivatives preserve the codomain harmonic fibers, so $v \in \mathcal{Q}_r$. Both p_π and t transform by sign on A_π, making v A_π-invariant. Hence $\sigma_r(e_\pi) = v$ defines an equivariant seed. The constant coefficient of $\overline{A}_k^U$ is the identity. With the final harmonic projection retained, Fischer adjunction gives

$$\langle t, \mathcal{A}_0(\sigma_r)s\rangle = [G : A_\pi]\langle t, p_\pi v\rangle = [G : A_\pi]\|v\|^2 > 0. \tag{S11.4}$$

The scalar sign block is therefore nonzero, proving the assertion. The polynomials and seed in this construction have rational coefficients. □

At $(a, b) = (6, 7)$ there is an explicit certificate independent of the ideal-containment theorem. Write the six codomain blocks as $(x_{1c}, \dots, x_{6c}, z_c)$, and take as domain blocks the six x-columns and the block $(z_1, \dots, z_6)$. Thus

$$p = \Delta_6(z)\prod\nolimits_{c=1}^{6} \Delta_6(x_{1c}, \dots, x_{6c}), \qquad q = \prod\nolimits_{c=1}^{6} \Delta_7(x_{1c}, \dots, x_{6c}, z_c)\Delta_6(\eta).$$

Put $K = \Delta_6(D)\Delta_6 = 6!\prod_{j=0}^{5} j!$. The derivative differences in each $\Delta_6(D_{x_c})$ annihilate every codomain block mean. Evaluating at $x = 0$ therefore gives

$$p(D)q|_{x=0} = K^6\, 7^{-15}\Delta_6(D_z)\big((z_1 \cdots z_6)^6\Delta_6(z)\big).$$

The coefficient of $z_1^6 \cdots z_6^6$ is

$$K^6\, 7^{-15}\frac{6!\prod_{j=0}^{5}(6 + j)!}{(6!)^6} > 0. \tag{S11.5}$$

Indeed, only matching permutations in the two alternating determinant expansions contribute, and their signs multiply to one. Translation invariance makes the derivative $p(D)q$ nonzero on W_0. The resulting seed has degree 36, from sign degree 105 to 141.

S11.4. **A degree condition on determinant certificates.** Before any rank computation, degrees alone restrict the seeds that a determinant certificate can use. For homogeneous generators M_i of shifts r_i, every nonzero monomial in $\det(\sum_i M_i \otimes Z_i)$ has weighted degree $d\delta_\lambda$, where each entry of $Z_i \in M_d(\mathbb{C})$ has weight r_i and

$$\delta_\lambda = \sum\nolimits_k k(\dim \mathcal{E}^Q_{\lambda,k} - \dim \mathcal{E}^P_{\lambda,k}) = [s_\lambda]F_{a,b}(X;1).$$

To check the weighted degree, sum codomain degree minus domain degree along a determinant term; each degree occurs d times. Proposition S7.1 already proves $\delta_\lambda \geq 0$ uniformly.

At $(6,7)$, writing $\chi_j = \chi^\lambda(2^j 1^{42-2j})$, the formula for δ_λ becomes

$$\delta_\lambda = \tfrac{3}{2}(5d_\lambda - 7\chi_1 + 7\chi_6 - 5\chi_7).$$

An exact character calculation over all 53 174 partitions of 42 finds equality only for the four neutral labels $(42), (41,1), (40,2), (40,1,1)$; the calculation also checks dimensions and character orthogonality. The constraint is a necessary condition on degrees; it computes no rank. More generally, seeds of shift zero alone have zero amplified quantum permanent on every nonneutral block, since an invertible degree-preserving map would force equality in every degree. Positive-shift seeds are therefore essential.

S11.5. **An exact certificate on the sign-twisted standard block.** The sign-twisted standard block at $(a,b) = (6,7)$ has multiplicity 41, so separate nonzero entries do not suffice, as they did on the sign block. Two homogeneous seeds give a common nonsingular matrix.

Proposition S11.3 (A scalar certificate beyond sign)**.** *At $(a,b) = (6,7)$, the current space (S9.1) on $\lambda = (2,1^{40})$ contains an invertible matrix. Two currents with $j = 0$, of seed degrees 30 and 24, suffice. Thus the original current space on this block satisfies (S9.2), and its scalar quantum permanent is strictly positive.*

Proof. ***The two seeds.*** Both seeds come from products of S_6-invariant harmonics, one factor for each codomain block. Use the domain and codomain block partitions in the explicit sign certificate above. Their stabilizers $A, B \subseteq S_{42}$ have intersection $H = (S_6)^6 \rtimes S_6$, with $[A:H] = 5040$. In a centered seven-variable block, write ξ for its distinguished coordinate and p_j for the sum of the jth powers of all seven coordinates. The following S_6-invariant polynomials are harmonic:

$$\psi_4(\xi) = \xi^4 - \tfrac{3}{7}p_2\xi^2 - \tfrac{2}{7}p_3\xi + \tfrac{3}{49}p_2^2 - \tfrac{1}{7}p_4,$$
$$\psi_5(\xi) = \xi^5 - \tfrac{33}{70}p_2\xi^3 - \tfrac{23}{70}p_3\xi^2 + (\tfrac{3}{28}p_2^2 - \tfrac{33}{140}p_4)\xi + \tfrac{4}{35}p_2p_3 - \tfrac{1}{7}p_5.$$

Indeed, applying the symmetric constant-coefficient operators of degrees two through five gives zero; translation invariance supplies the degree-one equation. For the codomain blocks $K_c = (x_{1c}, \dots, x_{6c}, z_c)$ with means η_c, set

$$f_{30} = \prod\nolimits_{c=1}^6 \psi_5(z_c - \eta_c), \qquad f_{24} = \prod\nolimits_{c=1}^6 \psi_4(z_c - \eta_c). \tag{S11.6}$$

Each factor uses the centered coordinates of its own block in the power sums. Both polynomials are H-invariant harmonics in the codomain fiber, with outer degree zero. For $r = 30, 24$, the vector $\sum_{g \in A/H} g(f_r \otimes e_\kappa)$ is A-invariant in $\mathcal{Q}_r$, and hence defines an equivariant seed $U \to \mathcal{Q}_r$. Let M_r be its $j = 0$ current, retaining the final harmonic projection.

Bases. To compute the matrices of M_{30} and M_{24} exactly, we fix bases of the two multiplicity spaces. Put $\Delta = \Delta_6$ and $K = \Delta_6(D)\Delta_6 = 6! \prod_{j=0}^5 j!$. For the seven centered domain block means, let $\phi_t(s_R)$ be the S_6-invariant harmonic of degree t, normalized to have coefficient one on s_R^t in its expression in $s_R, p_2, \dots, p_t$; put $\phi_0 = 1$. The domain basis consists of inner labels (j,t), $1 \leq j \leq 5$, $0 \leq t \leq 6$, of degree $105 - j + t$, and outer labels $t = 1, \dots, 6$ of degree $105 + t$. For a coordinate i

in domain block R, their fiber covariants are

$$p_{(j,t),i} = \Big(\prod\nolimits_{R'\neq R} \Delta_6(w_{R'})\Big)\,(D_i^j\Delta_6(w_R))\phi_t(s_R),$$
$$p_{t,i} = \Big(\prod\nolimits_{R} \Delta_6(w_{R})\Big)\,\phi_t(s_R).$$

The codomain basis has inner labels (h,v), $1\le h\le 6$, $0\le v\le 5$, of degree $141-h-v$, and outer labels $v=1,\dots,5$ of degree $141-v$. For $i\in K_c$, use

$$q_{(h,v),i} = \Big(\prod\nolimits_{K'\neq K_c} \Delta_7(w_{K'})\Big)\,(D_i^h\Delta_7(w_{K_c}))D_c^v\Delta(\eta),$$
$$q_{v,i} = \Big(\prod\nolimits_{K} \Delta_7(w_{K})\Big)\,D_c^v\Delta(\eta).$$

The inner derivatives are centered block derivatives; on a Vandermonde they equal ordinary coordinate derivatives. These bases follow by restricting the ambient standard representation to inner centered coordinates and block means. The marked-block outer representation is the permutation representation, with one harmonic copy in each degree $0,\dots,6$ on the domain and its sign twist in degrees $15-v$ on the codomain. Distinct labels of the same total degree have distinct inner and outer degrees, so their Fischer Gram factors are nonzero and diagonal.

Matrix entries. We express each matrix entry through one contraction of basis covariants. Both coordinate families sum to zero, so the trace contraction on the standard representation is

$$R_{\beta\alpha} = \sum\nolimits_{i=1}^{42} p_{\alpha,i}(D)q_{\beta,i} \tag{S11.7}$$

for every pair of labels α,β. Each $R_{\beta\alpha}$ is an H-invariant codomain harmonic of the required seed degree. By Fischer adjunction, its pairing with f_r equals the current matrix entry, up to nonzero row and column Gram factors and a common factor. To check the common factor, sum first over the cosets G/A and then over the cosets A/H that define the seed. Every summand equals the pairing of the seed with (S11.7), so the common factor is $[G:A][A:H]=[G:H]$. The harmonic projection disappears only inside this pairing with harmonic codomain covariants.

Set $x_{ic}=y_c$, $u_c=z_c-y_c$, and then $\eta_c=(6y_c+z_c)/7=0$ in (S11.7). The resulting polynomials are symmetric in $u_1,\dots,u_6$. Up to nonzero scalar factors, pairing with f_{30} and with f_{24} is given by the two coefficient functionals

$$[u_1^5\cdots u_6^5]R_{\beta\alpha}, \qquad [u_1^4\cdots u_6^4]R_{\beta\alpha} \tag{S11.8}$$

respectively. In fact, setting $\eta=0$ retains only outer degree zero. Each inner S_6-invariant harmonic space has one generator in every degree $0,\dots,6$, and the displayed specialization gives

$$\psi_4 = \tfrac{30}{343}u^4, \qquad \psi_5 = \tfrac{3}{245}u^5.$$

Orthogonality of the separate inner degrees then proves the claim about the two functionals. The same argument shows that they define single common seeds across all domain degrees, rather than independently chosen entries.

The calculation can be performed in six variables. One first uses

$$\sum\nolimits_{i=1}^{6}(\partial_i^j\Delta_6)(D_x)\partial_i^h\Delta_7(x,z)\Big|_{x_i=y} = Kc_{jh}(z-y)^{6+j-h},$$

where, for $1\le j\le 5$ and $1\le h\le 6$,

$$c_{jh}=0\ (h<j), \qquad c_{jj}=\frac{6^{\underline{j+1}}}{j+1}, \qquad c_{jh}=-6^{\underline{h}}\ (h>j).$$

The unmarked contraction is $K(z-y)^6$, and differentiating its distinguished z coordinate h times gives $K6^{\underline{h}}(z-y)^{6-h}$. These identities follow by expanding both Vandermonde determinants and matching exponent lists. An off-diagonal contribution replaces one exponent from $0,\dots,5$ by 6,

with one sign reversal. The remaining outer contraction uses, for A homogeneous of degree $15-v$ and $\zeta=(-5,-3,-1,1,3,5)$,

$$\langle D_c^v\Delta, A\rangle = 2^{-15}\sum\nolimits_{\tau\in S_6}\operatorname{sgn}(\tau)\zeta_{\tau,c}^v A(\zeta_\tau). \tag{S11.9}$$

Indeed, alternation in degree fifteen is a scalar multiple of Δ, and $\Delta(\zeta)=2^{15}\prod_{j=0}^5 j!$.

The determinant modulo a prime. Nonsingularity over $\mathbb{Q}$ follows from nonsingularity modulo a suitable prime. After removing the common nonzero row-column factor $K^6 7^{v-15}6^{-t}$ from each pairing, reduce the rational entries modulo the prime 1000033. The degree-thirty matrix has rank forty. Every block $k\to k+30$, $100\le k\le 110$, has maximal column rank; the only unused domain line is degree 111, and the codomain cokernel lies in degree 135. Append the column of M_{24} at domain degree 111 to the $105\to 135$ block. The resulting square blocks have these determinants.

Source k	Size	Determinant	Source k	Size	Determinant
100	1	969758	106	6	717690
101	2	272283	107	5	429366
102	3	847429	108	4	26984
103	4	945704	109	3	655031
104	5	52676	110	2	91829
105,111	6	824074			

In the basis order (j,t) lexicographically followed by outer t, and (h,v) lexicographically followed by outer v, the combined 41×41 determinant is

$$346965 \pmod{1000033} \neq 0. \tag{S11.10}$$

The prime avoids every denominator in the harmonic coefficients and the displayed calculation, so the combined matrix is nonsingular over $\mathbb{Q}$. The coefficient construction and a second implementation using sparse polynomial arithmetic independently give all 181 needed entries and (S11.10).

Back to the current space. The combined matrix mixes columns of two currents, so a degree count is needed to show that its determinant certifies the original current space. The difference of the sums of codomain and domain degrees is 1224. A determinant term with m shift-24 entries and $41-m$ shift-30 entries must satisfy $1230-6m=1224$, hence $m=1$. Since the shift-30 map kills domain degree 111, that one entry must be its $111\to 135$ column. All other shift-24 columns are irrelevant to the determinant. Rescaling the two coefficient functionals to (S11.6) only multiplies it by a nonzero factor. Therefore $M_{30}+M_{24}$ is invertible. More precisely,

$$\det(xM_{30}+yM_{24}) = cx^{40}y, \qquad c\neq 0.$$

The CP map with these two Kraus matrices has quantum permanent $40!|c|^2>0$, proving the assertion. □

The grading calculation has a uniform extension, but the rank certificate above is specific to $b=7$. For $a=6$ and $b\ge 6$, the domain and codomain sign-twisted standard bases are indexed by $\{0,\dots,5\}\times\{0,\dots,b-1\}\setminus\{(0,0)\}$, with degrees

$$15b-x+y, \qquad 6\binom{b}{2}+15\varepsilon_b+(-1)^{\varepsilon_b}x-y,$$

respectively. Reflect y to $b-1-y$, and also reflect x to $5-x$ when b is even. Every index except the one reflected to $(0,0)$ matches its reflection with shift

$$R=3b^2-19b+6+10\varepsilon_b;$$

the exceptional index is matched with itself, with shift $R-(b-6+5\varepsilon_b)$. Both shifts are nonnegative for $b\ge 6$; at $b=7$ they are 30 and 24. This degree matching is feasible, but it proves nothing about the rank of the currents. Moreover, the minimum codomain degree minus the maximum domain

degree is $3b^2 - 20b + 2 + 10\varepsilon_b$. At $b = 8$ it is 34, so currents of shifts 24 and 30 vanish on this block. A uniform certificate must use seeds whose degrees grow with b.

S11.6. **Invertible blocks in the first positive degrees.** For labels of small depth, a current with seed degree zero is invertible in the first positive degrees, because the harmonic projection acts there as the identity. Deleting these rows and columns leaves smaller residual matrices to test.

Proposition S11.4 (Low-degree current minors)**.** *Let $a = 6$, $b \geq 6$, and $N = 6b$. For every $\lambda \vdash N$ of depth $N - \lambda_1 \leq 5$, a $j = 0$ current with seed degree zero is invertible from domain degree one to codomain degree one. On $\lambda = (N - 3, 1^3)$, such a current is also invertible between the degree-two multiplicity spaces, both of dimension two. After the ground columns are eliminated and each current is split into its degree blocks, deleting these matching rows and columns preserves the expansion problem on the remaining space. At $(6, 7)$, the residual matrices on the three depth-three labels are square, of sizes*

λ	$(39, 3)$	$(39, 2, 1)$	$(39, 1^3)$
residual size	10614	21277	10658.

We make no assertion about the maximal minors of these residual matrices.

Proof. The ground modules U and V have equal multiplicities through depth six. To see this equality, mark $r \leq 6$ labels and take invariants under S_{N-r}. For either wreath permutation module, the orbits correspond to the set partitions induced on the marked labels, and every set partition of the marked labels fits both block dimensions. The resulting S_r-permutation modules are therefore identical. Triangular branching recovers the equal ground multiplicities at each depth up to r.

Put $W = W_0^*$. There are no linear relations in either harmonic ideal, so $\mathcal{P}_1 = W \otimes U$ and $\mathcal{Q}_1 = W \otimes V$. Tensoring with W changes depth by at most one, by the restriction–induction rule for the standard representation. For a label of depth at most five, choose a ground seed $\mu : U \to V$ that is an isomorphism on every contributing ground isotype. The current in degree one is exactly $I_W \otimes \mu$, and its codomain harmonic projection is the identity, which proves the first assertion.

In degree two, the relation module has character

$$\operatorname{ch} \mathcal{R}_{a,b,2} = h_b[h_a] + h_a h_{b-1}[h_a].$$

The first summand is the outer quadratic relation, and the second comes from the permutation representation on the inner quadratic relations. These are independent because the inner difference spaces and the outer mean space are orthogonal. We apply the hook extraction identity

$$F[1 - u] = (1 - u) \sum\nolimits_{r=0}^{N-1} (-u)^r [s_{(N-r,1^r)}] F$$

to this relation character, which specializes to $(1 - u) + (1 - u)^2$ and therefore has no hooks with leg at least two. Thus on $(N - 3, 1^3)$ both degree-two harmonic isotypes equal the full free isotypes. Every contributing ground label has depth at most five, so $I_{\operatorname{Sym}^2 W} \otimes \mu$ is again an invertible current. The two contributing ground labels are $(N - 2, 2)$ and $(N - 4, 2, 2)$, each with ground and branching multiplicity one. These labels and multiplicities follow by two-box restriction, since $\operatorname{Sym}^2 W$ is the permutation representation on unordered pairs, and in compatible bases the current is $\operatorname{diag}(x, y)$, with determinant xy.

To justify deleting the matching rows and columns, split each current into its degree blocks, which are then independent. After deleting the ground columns, decompose a residual map by its first positive degree as

$$\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}.$$

The block space for A contains an invertible matrix. A shrunk subspace for D remains shrunk in the full space. Write $\operatorname{proj}_{\text{low}}$ for the projection onto the first positive degree and E_{later} for the sum of the later degrees. Conversely, for any domain subspace Z, the image under the A block

contributes $\dim \mathrm{proj}_{\mathrm{low}} Z$ dimensions, and expansion for D contributes at least $\dim(Z \cap E_{\mathrm{later}})$ in disjoint codomain degrees. Their sum is $\dim Z$. The two expansion tests are therefore equivalent, and the graded-witness lemma transfers this equivalence back to the original current space. The three total dimensions at $(6,7)$ are $10619, 21280, 10660$. Their ground dimensions are $1,0,0$, and their degree-one dimensions are $4,3,0$; the last block additionally loses its two degree-two rows and columns. These numbers give the displayed residual sizes. □

S12. Exact checks

The proof of Proposition S11.3 rests on a machine computation, so we record how it was checked. The computation consists of the rational harmonic construction, the modular coefficient matrices, and the determinant reconstruction. Two implementations agree on all 181 required entries. The first uses finite polynomial jets; the second uses sparse polynomial dictionaries and also checks the full 720-term outer alternation. Direct polynomial differentiation verifies the same construction at $(a,b) = (2,3)$. The nonvanishing modulo the prime certifies the $(6,7)$ block.

References

François Bergeron. A q-analog of Foulkes' conjecture. *The Electronic Journal of Combinatorics*, 24 (1):Paper 1.38, 2017. doi: 10.37236/6004.

Ankit Garg, Leonid Gurvits, Rafael Oliveira, and Avi Wigderson. Operator scaling: Theory and applications. *Foundations of Computational Mathematics*, 20:223–290, 2020. doi: 10.1007/s10208-019-09417-z. URL https://www.math.ias.edu/~avi/PUBLICATIONS/GargGOW20.pdf.

Leonid Gurvits. Classical complexity and quantum entanglement. *Journal of Computer and System Sciences*, 69(3):448–484, 2004. doi: 10.1016/j.jcss.2004.06.003. Numbered locators refer to the preprint *Classical deterministic complexity of Edmonds' problem and quantum entanglement*, https://arxiv.org/abs/quant-ph/0303055v1.

Mark Haiman. Hilbert schemes, polygraphs and the Macdonald positivity conjecture. *Journal of the American Mathematical Society*, 14(4):941–1006, 2001. doi: 10.1090/S0894-0347-01-00373-3.

Ian G. Macdonald. *Symmetric Functions and Hall Polynomials*. Oxford University Press, Oxford, 2nd edition, 1995.

Ivan Marin. L'algèbre de Lie des transpositions. *Journal of Algebra*, 310:742–774, 2007. doi: 10.1016/j.jalgebra.2006.09.029. URL https://arxiv.org/abs/math/0502119.

Philippe Moustrou, Cordian Riener, and Hugues Verdure. Symmetric ideals, Specht polynomials and solutions to symmetric systems of equations. *Journal of Symbolic Computation*, 107:106–121, 2021. doi: 10.1016/j.jsc.2021.02.002. URL https://arxiv.org/abs/1912.05266.

OpenAI. Foulkes' conjecture for the sixth symmetric power, 2026. URL https://www.emergentmind.com/openai-math-explorer/papers/373.pdf. OpenAI Math Explorer, manuscript 373, 25 September 2026.

Alexios P. Polychronakos. Exact spectrum of SU(n) spin chain with inverse-square exchange. *Nuclear Physics B*, 419:553–566, 1994. URL https://arxiv.org/abs/hep-th/9310095.

Suvrit Sra. The q-Foulkes problem I: A fusion inequality for Macdonald plethysms. Preprint, 2026a.

Suvrit Sra. The q-Foulkes problem II: Schur positivity with high probability. Preprint, 2026b.

Suvrit Sra. The q-Foulkes problem III: On Foulkes' conjecture with a graded inner factor. Preprint, 2026c.